Solar Compass 19 (2026) 100182

Contents lists available at ScienceDirect

Solar Compass

journal homepage: www.elsevier.com/locate/solcom

Explainable AI in solar PV forecasting: A systematic review of models, explainers, and physical drivers

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ARTICLE INFO

Keywords:
Solar energy
XAI
SHAP
Solar PV forecasting
Machine learning

ABSTRACT

Solar photovoltaic (PV) forecasting supports secure grid operation, yet advanced machine learning models are hard to trust when their decisions are opaque. This PRISMA 2020 guided systematic review examines explainable artificial intelligence (XAI) in solar forecasting. Of 951 retrieved records, 173 met strict eligibility, requiring quantitative forecasting evaluation plus an explicit explanation output. A sensitivity analysis of the decisive screening rule, which excluded 299 records using attention only as an architectural component, supported its precision, and dual independent re-screening of 75 excluded records found no false exclusions. Long short-term memory (LSTM) models dominated (48 studies), ahead of transformers (39) and convolutional neural networks (CNN, 32) used mainly for image-based nowcasting. SHAP prevailed among explainers (69 studies), ahead of feature importance and permutation methods (39), attention visualization (31), and LIME (14), and 47 studies combined several methods. Quality-stratified synthesis showed these rankings are robust, with the ordering unchanged among the 114 high-quality studies. Irradiance ranked first across targets, while temporal encodings and ambient temperature were stable secondary contributors. Humidity, wind, and cloud proxies shifted with season and climate zone, making global rankings conditional. Horizon mattered, with longer horizons using tabular weather and post hoc attribution and ultra-short horizons using sky images. Cross-study benchmarking remains fragile, and quantitative evaluation of explanation fidelity, stability, and runtime is rare. A critical appraisal of the dominant explainers documents computational cost, instability, sensitivity to correlated inputs, and unverified faithfulness, alongside a reproducibility gap, with only 1.7% of studies releasing both code and data.

1. Introduction

1.1. Background and context

Solar photovoltaic (PV) power is one of the fastest growing electricity sources, and its expansion is accelerating across utility scale plants and distributed rooftops. The International Energy Agency (IEA) projects that solar PV will deliver the largest share of new electricity capacity additions through 2030, enabled by falling costs, policy support, and stronger decarbonization mandates. Yet PV output is tightly coupled to irradiance variability, cloud fields, aerosols, and local weather regimes, so the same installed capacity can yield very different power trajectories across minutes, hours, and days. This variability complicates dispatch, reserves, and market participation, making accurate PV forecasting a core operational need. Reliable characterization of the solar resource also underpins solar technologies beyond grid connected PV. Recent work applies artificial intelligence and optimization to hybrid photovoltaic/thermal (PV/T) collector design [1,2], to hydrogen oriented and thermoelectric PV/T configurations [3], and to techno-economic assessment of PV/T systems for residential supply [4]. These system level applications depend on trustworthy solar forecasts, which reinforces the case for transparent, explainable forecasting models.

Data driven learning has become the dominant approach, and it frequently outperforms physical and statistical baselines in comparative studies [5,6]. Widely used pipelines span classical learners and deep networks, with support vector regression (SVR) and tree ensembles still common when data are moderate [7]. Long short-term memory (LSTM) forecasting remains prominent for temporal structure [8,9], convolutional neural network (CNN) based models support spatial and image inputs [10], and attention based Transformers are increasingly reported for multi input dependence [11]. As model depth increases, interpretability often decreases, creating a practical trade off that limits adoption.

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https://doi.org/10.1016/j.solcom.2026.100182
Received 8 April 2026; Received in revised form 19 July 2026; Accepted 21 July 2026
Available online 30 July 2026
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Table 1
Discriminative comparison of recent AI solar forecasting reviews.

Criterion	Our review	[14]	[15]	[16]
Explicit study selection rules and exclusions	✓	✓	✓	✓
Deep learning architectures comprehensively mapped	✓	✓	✓	⊙
Primary scope is Explainable AI (XAI) and interpretability	✓	×	×	×
Synthesizes feature importance consensus from model outputs	✓	×	×	×
Critiques the absence of quantitative XAI evaluation metrics	✓	×	×	×
Primary scope is Hybrid Physics + ML integration	×	×	×	✓
Synthetic data or Generative AI is the primary focus	×	✓	⊙	×
Bibliometric geographic analysis with country-level mapping	✓	×	×	✓
Benchmarking critique (unnormalized metrics & evaluation gaps)	✓	×	⊙	⊙
Number of papers reviewed	173	84	26	67

Notes ✓ clearly present as a substantive component, ⊙ present but limited, × not present.

This black box behavior is problematic in regulation sensitive and safety critical energy settings. Operators need forecasts that are both accurate and explainable, including which inputs were influential and whether the model is stable under changing weather regimes [12]. Similar trust barriers have been documented in energy AI deployments more generally [13].

1.2. Comparison with other reviews

Recent surveys map AI progress in solar forecasting, yet interpretability is usually treated as a secondary theme rather than a synthesis objective. Table 1 positions this review relative to close survey neighbors, and it clarifies why a dedicated XAI focused synthesis is still needed.

The review in [14] prioritizes generative AI for forecasting and system optimization, emphasizing augmentation and scenario generation. The systematic review in [15] organizes deep learning forecasting primarily around accuracy, preprocessing, and architectural families. The taxonomy in [16] is strongest on hybrid physics plus learning workflows, and it links transparency mainly through physical constraints instead of formal XAI evaluation. Across these works, explainers such as SHapley Additive exPlanations (SHAP), Local Interpretable Model-agnostic Explanations (LIME), and partial dependence analysis are not systematically audited, and explanation quality is rarely assessed with explicit criteria.

This review follows PRISMA 2020 and treats interpretability as reported evidence. It aggregates XAI oriented studies to show how feature importance is computed and communicated, and whether explanations match known meteorological drivers of PV variability. A key gap emerges, explanations are often presented, but their reliability is seldom tested, such as faithfulness, stability, and runtime suitability for near real time use.

1.3. Machine learning for solar PV forecasting

ML and DL adoption in solar and microgrid research has grown rapidly [17–19], supported by broader renewable energy analytics work [20–22]. PV forecasting is a major sub domain within this literature [23,24]. Traditional ML models remain common, including SVR [5,25,26] and Random Forest [6,27–29]. Gradient boosting methods are also widely reported for strong tabular performance [27,30–32], and they often remain competitive under realistic data constraints [33,34]. These baselines can be relatively interpretable and computationally efficient, yet they often struggle with long horizon temporal dependence and multiscale variability, especially when spatial signals matter [35,36].

Deep learning improved representation learning for time and space. LSTM and BiLSTM models are widely used for sequence prediction [8,30,31,37], while CNNs ingest sky imagery and satellite inputs [38–41]. Hybrid CNN LSTM designs combine both structures for end to end spatiotemporal learning [42,43]. More recently, attention based Transformers, including TFT style models, have shown competitive results [44–47]. Their adoption has accelerated with richer datasets and longer training regimes [48–50]. Across the 173 studies synthesized here, LSTM families are the most frequently used, with feedforward baselines, Transformers, tree ensembles, and CNN or TCN variants all strongly represented, and Transformer uptake rising sharply after 2023. Still, many of the strongest models remain hard to interpret, and explanation is often added late or reported lightly [51].

1.4. Explainable Artificial Intelligence (XAI)

Explainable Artificial Intelligence (XAI) includes methods that make model outputs and internal behavior more understandable, a need sharpened by regulatory expectations and by the wider push for accountable AI [52]. In forecasting, explanations support trust, debugging, and monitoring of dataset shift, so they matter even when compliance is not the main driver.

XAI methods are commonly grouped as ante hoc versus post hoc, local versus global, and model agnostic versus model specific. In PV forecasting, post hoc feature attribution is the most common approach. SHAP is frequently used to produce additive feature contributions [53–55], and it is applied across PV studies for ranking meteorological drivers [6,56]. Temporal and model wide attributions are also reported for sequence models [57,58]. LIME is common for local surrogate explanations [59–61], and it is often used for case based narratives and local sensitivity reports [6,28]. Permutation feature importance (PFI) is used as a simpler ranking tool [8,27], sometimes paired with partial dependence plot (PDP) style plots for average effect communication [62,63]. For image based models, gradient-weighted class activation mapping (Grad-CAM) and integrated gradients appear in selected studies [38,39]. A smaller but growing line of work uses inherently interpretable architectures, embedding transparency directly in the model mathematics [9,64,65].

A critical methodological point in this review is the attention distinction. Attention weights can look explanatory, but they are not automatically faithful [66,67], and many papers use attention only to boost accuracy. We therefore separate attention as architecture from attention as explanation, and we exclude 299 studies that did not analyze attention weights as explainers. Because this is the most consequential screening decision in the review, Section 2.6 reports a dedicated sensitivity analysis of this rule, including an independent re-screening of a random sample of the excluded records.

1.5. Motivation and gap analysis

Despite fast publication growth, evidence on XAI for PV forecasting is scattered and reporting practices are inconsistent. Broad energy XAI reviews blend solar with other targets, while solar AI reviews often mix forecasting with fault diagnosis, MPPT, sizing, and control, blurring PV forecasting specific explanation needs. Many XAI mentions remain aspirational, with limited evaluation of fidelity, stability, and computational cost.

This review provides a PV focused synthesis that audits XAI usage and explanation reporting, while enforcing a strict attention screening rule. The pace of work, with 43 studies in 2024 and 73 in 2025, makes a structured consolidation timely.

1.6. Objectives and research questions

The objective of this systematic review is to identify, evaluate, and synthesize literature on XAI applied to machine learning based solar PV generation forecasting.

1. RQ1 What ML and DL architectures are most commonly used for PV forecasting with XAI, and which achieve the best predictive performance
2. RQ2 Which XAI techniques have been applied, and how are explanations reported and validated in practice, including evidence on fidelity, stability, and runtime
3. RQ3 What input features are most influential according to XAI analysis, and do findings converge across studies
4. RQ4 How does XAI application vary across forecasting horizons
5. RQ5 What datasets, geographic regions, and climatic zones are represented, and what gaps exist
6. RQ6 What challenges, limitations, and open research directions remain for XAI in PV forecasting

1.7. Contributions of this review

This review contributes the following.

- A PRISMA 2020 guided systematic review of XAI for PV forecasting, covering 173 studies from 2016 to 2026
- An attention architecture screening rule that separates attention as architecture from attention as explanation, excluding 299 records, together with a quantitative sensitivity analysis of this rule based on dual independent re-screening of a random sample of the excluded records
- A dual axis taxonomy mapping ML families and XAI categories, enabling structured field interpretation
- Aggregated feature importance consensus across XAI studies, highlighting consistently influential PV predictors
- A 10 item quality assessment framework for XAI in energy forecasting, with scored evaluations for all included studies, and a quality-stratified synthesis showing which findings persist when the corpus is restricted to high-quality studies
- A cross-study evidence synthesis that grades the consistency of the main findings by the number of independent supporting studies and their quality tiers
- A critical appraisal of the dominant XAI techniques covering computational complexity, explanation instability, sensitivity to correlated inputs, faithfulness, and reproducibility
- A research roadmap outlining gaps in standardized metrics, real time deployment constraints, reproducibility infrastructure, and geographic diversity

1.8. Paper organization

Section 2 describes the methodology workflow and synthesis steps for RQ1 through RQ6. Section 3 reports results on the study flow, bibliometrics, quality outcomes, taxonomy mapping, and research question findings. Section 4 discusses implications and limitations, and Section 5 closes with summary and future directions.

2. Methodology

This section explains the full review process used to find, screen, extract, and synthesize evidence on explainable AI for solar forecasting. Because many papers describe a model as explainable without showing any explanation outputs, the methodology prioritizes verifying that XAI was actually implemented, and that target variables are defined consistently. The subsections outline the eligibility rules, information sources, database queries, screening stages, extraction fields, quality scoring, and the synthesis steps used to answer RQ1 through RQ6.

2.1. Review methodology framework

This review followed a structured evidence mapping workflow designed for a common problem in this literature: many papers claim explainability but do not show explanation outputs. The workflow was defined before screening began, then applied consistently to all records, to reduce ad hoc inclusion decisions and limit subjective drift during screening. In this review, explainability is treated as an implemented analysis with interpretable outputs, not as a model label, so attention based models were included only when the authors analyzed attention weights as explanations.

In practice, the framework follows sequential stages that link literature retrieval to synthesis. First, the search strings were built around three concept blocks, interpretability, learning based forecasting, and solar or PV targets, then adapted to each database constraint while maintaining the same conceptual scope. Second, all retrieved records were consolidated and deduplicated using DOI matching and fuzzy title matching. Third, title and abstract screening applied the inclusion and exclusion criteria, with a strict boundary between attention as architecture and attention as explanation. Fourth, full texts were checked to verify genuine XAI execution and the reporting of quantitative forecasting metrics. Fifth, all included studies were extracted using a standardized 15 field form covering model family, XAI method, horizon, inputs, dataset descriptors, evaluation metrics, and the main explanation findings. Sixth, study quality was assessed using a 10 item checklist to weight the interpretation of evidence rather than to filter studies out. Finally, synthesis combined taxonomy mapping, bibliometric trend analysis, and narrative integration aligned to the research questions. Screening decisions, extraction entries, and quality scores were recorded throughout, and can be made available for editorial audit if requested.

2.2. Target variable definitions and harmonization

The included literature forecasts related but non equivalent targets, therefore target definitions were harmonized during extraction and synthesis. In this review, *solar irradiance* refers to instantaneous radiative power per unit area, typically reported in W/m^2, including Global Horizontal Irradiance (GHI), Direct Normal Irradiance (DNI), and Diffuse Horizontal Irradiance (DHI). *Solar radiation* refers to time integrated energy over an interval, typically reported as MJ/m^2 per day or kWh/m^2 per day, and appears in studies estimating daily or monthly aggregates. *PV power output* refers to plant electrical power, typically in W, kW, or MW at a time step. *PV energy* refers to integrated electrical energy, typically in kWh, and is treated as a separate target when reported.

To avoid mixing fundamentally different physics and units, synthesis and benchmarking were performed within these target classes, and cross class performance claims were avoided unless a study reported a normalized metric with a clearly stated denominator.

2.3. Eligibility criteria

Studies were evaluated against six inclusion criteria (IC1–IC6) and seven exclusion criteria (EC1–EC7). Inclusion criteria are summarized in Table 2, and exclusion criteria are summarized in Table 3. A study must satisfy all inclusion criteria and none of the exclusion criteria to be included in the review.

Table 2
Inclusion criteria.

ID	Criterion
IC1	Explicitly implements at least one XAI method and reports its outputs for a solar PV forecasting model, examples include SHAP, LIME, Grad-CAM, Integrated Gradients, PFI, PDP/ICE, counterfactual explanations, attention weight visualization with interpretation, or inherently interpretable models with transparent structure.
IC2	Uses at least one ML, DL, or hybrid forecasting model targeting solar PV power, PV energy, irradiance, or PV generation, at any prediction horizon, provided the task is clearly framed as forecasting or prediction from historical or exogenous inputs.
IC3	Published as a peer reviewed journal article, peer reviewed conference paper, or peer reviewed review article, ensuring minimum methodological screening and archival stability.
IC4	Published within the predefined window January 2016 to February 2026, aligning the corpus with the modern period of widely adopted XAI frameworks and standardized reporting practices.
IC5	Written in English with accessible full text, sufficient to extract the modeling pipeline, data context, explainability procedure, and evaluation results.
IC6	Reports at least one quantitative forecasting performance metric, for example RMSE, MAE, R^2, MAPE, nRMSE, or an equivalent numeric error or skill metric, reported on a validation or test setting so comparisons remain meaningful.

2.3.1. Inclusion criteria

Six inclusion criteria (IC1-IC6) were defined to retain studies that both, perform solar PV forecasting, and report an implemented explainability analysis with verifiable outputs. A paper was included only when it satisfied all criteria.

IC1 requires particular care in operational screening. In this review, attention mechanisms or sensitivity style components only count as XAI when the authors actually analyze them as explanations, for example attention maps interpreted as feature or time step relevance, or saliency patterns discussed against physical drivers. If attention is used only to boost predictive performance, with no interpretability analysis of weights or attribution outputs, the study is excluded under EC2, even if the model architecture contains attention blocks.

2.3.2. Exclusion criteria

Seven exclusion criteria EC1 to EC7 were specified so the final corpus reflects authentic, implemented explainable AI applied to solar PV forecasting. Any paper triggering even one exclusion criterion was removed, even if it otherwise aligned strongly with the inclusion criteria.

EC1 removes studies that present solar PV forecasting models without applying any XAI method. In particular, papers that position explainability as a motivation, or a future extension, yet never execute an explainability technique, were excluded. EC2 is the most decisive filter in this review. Studies were excluded when they used attention modules or sensitivity analysis only as architectural elements or validation aids, while providing no interpretability analysis of the learned weights, saliency patterns, or derived explanations. The reasoning and the practical boundary for this decision are elaborated in Section 2.3.1 and also in Section 1.4. EC3 excludes studies whose central problem is not solar PV forecasting, including wind energy prediction, load forecasting, building energy modeling, or fault diagnosis, where solar variables appear only as secondary context.

EC4 excludes work that has not undergone peer review such as preprints, technical reports, and grey literature, to maintain a consistent baseline of methodological vetting. EC5 removes studies outside the defined period from January 2016 to February 2026, since work prior to 2016 largely predates the widespread adoption of modern, reusable XAI toolkits such as SHAP and LIME. EC6 excludes papers where XAI appears only as a superficial mention, for instance one sentence in discussion, or a suggested future direction, with no reported explainability outputs or analysis. EC7 excludes studies that report no quantitative forecasting metric. Without at least one numerical performance indicator, synthesis becomes fragile, and comparisons across research questions become unreliable.

Table 3
Exclusion criteria.

ID	Criterion
EC1	Does not apply any XAI technique to a solar PV forecasting model.
EC2	Uses attention or sensitivity analysis solely as an architectural component without interpretability analysis.
EC3	Not focused on solar PV forecasting.
EC4	Not peer-reviewed.
EC5	Outside publication date range.
EC6	XAI mentioned only superficially or as future work.
EC7	No quantitative performance metrics reported.

2.4. Search strategy

The search strategy was built around three conceptual domains, explainability or interpretability, learning based forecasting, and solar targets spanning PV power output, irradiance components, and radiation aggregates. Boolean AND logic was used between domains to ensure each retrieved record lies at their intersection, while Boolean OR logic was used within each domain to capture synonyms and related terms. Because each database imposes different query syntax, field tags, and query length limits, the query was adapted per source while preserving the same conceptual scope. All searches were restricted to English language publications from 2016 to 2026. Because author terminology varies widely, target classes were standardized during extraction using explicit unit based definitions.

Scopus and Web of Science supported the most comprehensive strings. IEEE Xplore restricts Boolean complexity, so the query was condensed and executed through the Command Search interface using the `All Metadata` field. ScienceDirect imposes strict length limits, therefore a shortened title, abstract, keywords query was used.

The executed searches returned 951 total records, Scopus 479, Web of Science 360, IEEE Xplore 72, and ScienceDirect 40, matching the identification totals in Fig. 2.

2.5. Study selection process

The study selection followed five phases.

Phase 1 Identification. Searches yielded 951 records.

Phase 2 Deduplication. DOI matching and fuzzy title matching removed 334 duplicates, leaving 617 records.

Phase 3 Title and Abstract Screening. Of 617 records, 399 were excluded. The exclusion breakdown was EC2 attention used purely as architecture 299, EC1 no XAI technique 47, EC2 sensitivity analysis used as model validation rather than XAI 25, EC1 or EC2 insufficient criteria match 13, EC5 outside the date range 7, EC3 fault only studies without forecasting 6, EC1 non solar domains 2. A further 34 uncertain records were reviewed manually and excluded, leaving 184 full text articles assessed for eligibility.

Phase 4 Full Text Review. Full texts were assessed with emphasis on genuine XAI implementation. Eleven full text articles were excluded under EC6 for superficial XAI, resulting in 173 studies included for qualitative synthesis.

Phase 5 Snowballing. Backward and forward citation tracking was performed, any candidate studies were screened using the same criteria, the final included set remained 173.

Scopus query. The Scopus search was executed using the TITLE-ABS-KEY field.

```
TITLE-ABS-KEY (
( "explainable artificial intelligence" OR "explainable AI" OR "XAI" OR "interpretab*" OR
  "explainab*" OR "SHAP" OR "SHapley Additive exPlanation*" OR "Shapley" OR "LIME" OR
  "local interpretable model-agnostic" OR "Grad-CAM" OR "saliency" OR "feature importance" OR
  "feature attribution" OR "feature contribution" OR "feature ranking" OR "permutation importance" OR
  "attention mechanism" OR "attention weight" OR "partial dependence" OR "counterfactual" OR
  "glass-box" OR "white-box" OR "transparent model" OR "sensitivity analysis" )
AND
( "machine learning" OR "deep learning" OR "neural network*" OR "LSTM" OR "long short-term memory" OR
  "BiLSTM" OR "GRU" OR "CNN" OR "convolutional neural network*" OR "transformer" OR
  "temporal fusion transformer" OR "random forest" OR "XGBoost" OR "LightGBM" OR "CatBoost" OR
  "gradient boosting" OR "support vector" OR "SVR" OR "ensemble learning" OR "hybrid model" OR
  "data-driven" OR "MLP" )
AND
( "solar power forecast*" OR "solar energy forecast*" OR "solar forecast*" OR "PV power forecast*" OR
  "PV forecast*" OR "PV generation forecast*" OR "photovoltaic power forecast*" OR
  "photovoltaic forecast*" OR "solar irradiance forecast*" OR "solar radiation forecast*" OR
  "solar power predict*" OR "solar energy predict*" OR "PV power predict*" OR "PV predict*" OR
  "photovoltaic power predict*" OR "photovoltaic predict*" OR "solar irradiance predict*" OR
  "solar radiation predict*" )
)
AND PUBYEAR > 2015 AND PUBYEAR < 2027
AND ( LIMIT-TO ( LANGUAGE , "English" ) )
```

Web of Science query. The Web of Science search was executed using the TS topic field.

```
TS=(
( "explainable artificial intelligence" OR "explainable AI" OR "XAI" OR "interpretab*" OR
  "explainab*" OR "SHAP" OR "SHapley Additive exPlanation*" OR "Shapley" OR "LIME" OR
  "local interpretable model-agnostic" OR "Grad-CAM" OR "saliency" OR "feature importance" OR
  "feature attribution" OR "feature contribution" OR "feature ranking" OR "permutation importance" OR
  "attention mechanism" OR "attention weight" OR "partial dependence" OR "counterfactual" OR
  "glass-box" OR "white-box" OR "transparent model" OR "sensitivity analysis" )
AND
( "machine learning" OR "deep learning" OR "neural network*" OR "LSTM" OR "long short-term memory" OR
  "BiLSTM" OR "GRU" OR "CNN" OR "convolutional neural network*" OR "transformer" OR
  "temporal fusion transformer" OR "random forest" OR "XGBoost" OR "LightGBM" OR "CatBoost" OR
  "gradient boosting" OR "support vector" OR "SVR" OR "ensemble learning" OR "hybrid model" OR
  "data-driven" OR "MLP" )
AND
( "solar power forecast*" OR "solar energy forecast*" OR "solar forecast*" OR "PV power forecast*" OR
  "PV forecast*" OR "PV generation forecast*" OR "photovoltaic power forecast*" OR
  "photovoltaic forecast*" OR "solar irradiance forecast*" OR "solar radiation forecast*" OR
  "solar power predict*" OR "solar energy predict*" OR "PV power predict*" OR "PV predict*" OR
  "photovoltaic power predict*" OR "photovoltaic predict*" OR "solar irradiance predict*" OR
  "solar radiation predict*" )
)
AND LA=(English) AND PY=(2016-2026)
```

IEEE Xplore query. IEEE Xplore was searched using Command Search with the All Metadata field.

```
("All Metadata":"explainable AI" OR "All Metadata":"XAI" OR "All Metadata":"interpretable" OR
 "All Metadata":"SHAP" OR "All Metadata":"LIME" OR "All Metadata":"feature importance" OR
 "All Metadata":"Grad-CAM" OR "All Metadata":"attention mechanism" OR "All Metadata":"explainability" OR
 "All Metadata":"Shapley" OR "All Metadata":"permutation importance" OR
 "All Metadata":"sensitivity analysis" OR "All Metadata":"feature attribution" OR "All Metadata":"saliency")
AND
("All Metadata":"machine learning" OR "All Metadata":"deep learning" OR "All Metadata":"neural network" OR
 "All Metadata":"LSTM" OR "All Metadata":"CNN" OR "All Metadata":"transformer" OR
 "All Metadata":"random forest" OR "All Metadata":"XGBoost" OR "All Metadata":"support vector" OR
 "All Metadata":"ensemble learning" OR "All Metadata":"gradient boosting" OR "All Metadata":"GRU" OR
 "All Metadata":"hybrid model")
AND
("All Metadata":"solar forecast" OR "All Metadata":"PV forecast" OR "All Metadata":"photovoltaic forecast" OR
 "All Metadata":"solar irradiance forecast" OR "All Metadata":"solar predict" OR "All Metadata":"PV predict" OR
 "All Metadata":"photovoltaic predict" OR "All Metadata":"solar power generation" OR "All Metadata":"solar energy forecast")
```

ScienceDirect query. ScienceDirect was searched over title, abstract, and keywords using a shortened string.

```
Title, abstract, keywords:
("explainable AI" OR "XAI" )
AND
(Solar AND ("forecast" OR "prediction" OR "predict" OR "Output" ))
```

2.6. Sensitivity analysis of the attention screening rule (EC2)

The exclusion of 299 records under EC2, attention used purely as an architectural component, is the most consequential screening decision in this review, removing 48.5% of the 617 deduplicated records. To test whether this rule could have distorted the evidence base, we performed a three part sensitivity analysis.

Part 1, composition audit of the excluded set. All 299 excluded records were profiled using their titles, abstracts, and indexed keywords. None of the 299 records mentioned any recognized explanation technique (SHAP, LIME, Grad-CAM, integrated gradients, PDP, ICE, PFI, saliency analysis, or counterfactual explanation) in the title, abstract, or keywords, whereas all 173 included studies were verified at full text to implement at least one. The excluded set is dominated by hybrid accuracy oriented deep learning: 215 of 299 records (71.9%) involve LSTM, GRU, or BiLSTM components, 120 (40.1%) involve CNN or TCN components, and 55 (18.4%) involve Transformer components, typically combined with signal decomposition and metaheuristic optimization. The temporal profile of the excluded records mirrors the included corpus, with 220 of 299 (73.6%) published in 2024–2025 (Fig. 1).

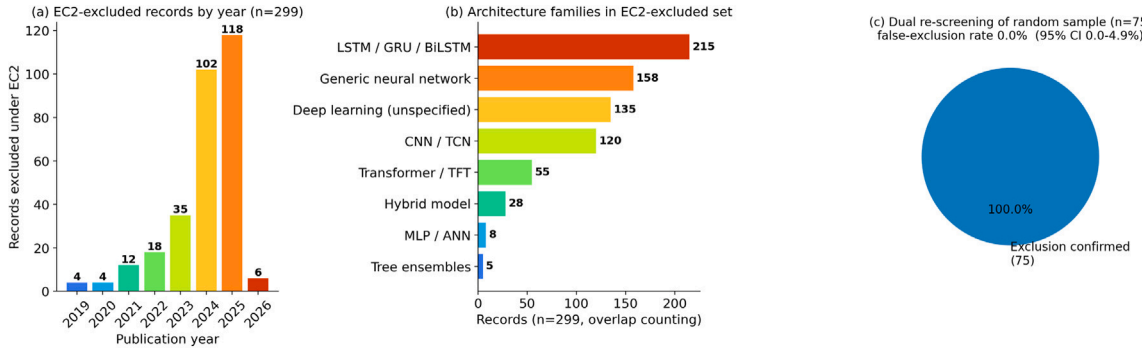


Fig. 1. Sensitivity analysis of the EC2 attention screening rule. (a) Publication years of the 299 excluded records. (b) Architecture families named in the excluded records (overlap counting). (c) Outcome of dual independent re-screening of a random sample of 75 excluded records.

Part 2, independent re-screening of a random sample. A simple random sample of 75 of the 299 excluded records (25.1%, fixed random seed 2026, sampled and analyzed in Python) was re-screened at title and abstract level by two independent screeners, one instructed to search actively for any indication of interpretability analysis, and one applying the criterion strictly as written. The two screeners agreed on 72 of 75 records (96% raw agreement); the three disagreements were resolved by a third adjudicator. The outcome was zero false exclusions in 75 records, corresponding to an estimated false exclusion rate of 0% with a Wilson 95% confidence interval of 0–4.9%. Even at the upper confidence bound, at most 15 of the 299 excluded records could plausibly merit inclusion, which would leave SHAP's leading position, the architecture landscape, and all thematic conclusions unchanged, and could at most shift the relative order of mid-ranked explainer categories (Section 3).

Part 3, concrete boundary examples. The re-screening produced archetypal examples of the pattern EC2 targets. Typical excluded designs include a BiGRU–TCN hybrid with dual-stage variational mode decomposition, where attention serves only to weight decomposed sub-series for accuracy; an LSTM–attention residual network for isolated microgrids evaluated solely on error metrics; and a self-attention EEMD–BiLSTM pipeline where attention weights spatiotemporal features but no weight is inspected, visualized, or discussed as an explanation. By contrast, included attention studies analyze the learned weights as evidence, for example inspecting horizon dependent shifts of attention from cloud imagery to clear sky irradiance features [68], or visualizing temporal attention heads of a temporal fusion transformer as variable importance [41,69].

If the 299 excluded records were counted as XAI studies solely because their architectures contain attention blocks, attention visualization would nominally become the dominant explanation category at approximately 70% of an enlarged corpus, despite none of these papers reporting an explanation output. The EC2 rule therefore preserves the focus of this review on implemented and reported explanations, and the conclusions are insensitive to plausible misclassification at the estimated rate.

2.7. Data extraction

This stage converts each included paper into a standardized record that supports consistent cross study comparison. Because extraction choices can silently bias a review, the process used a fixed codebook, repeat pass checks, and targeted verification of numeric fields and XAI claims, to minimize drift and transcription errors.

2.7.1. Extraction process

For the extraction we developed a predefined extraction form which contained 15 fields (D1-D15). These fields captured bibliographic metadata, model and XAI details, datasets, horizons, inputs, and reported

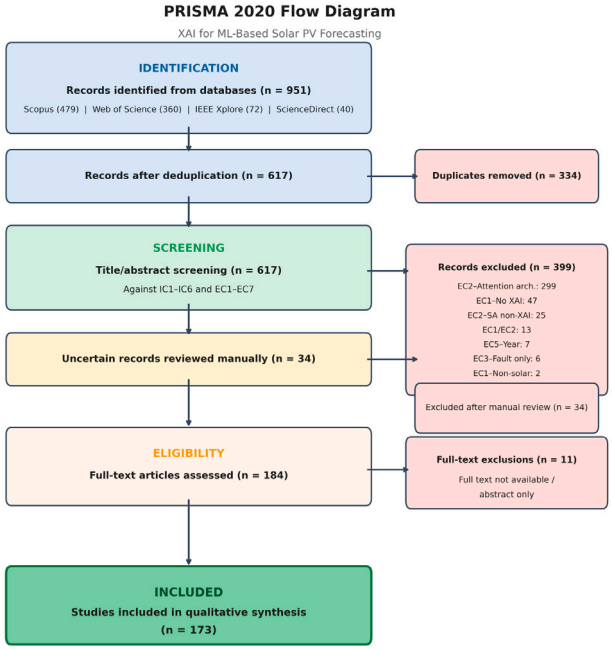


Fig. 2. PRISMA 2020 flow diagram for study selection.

metrics. Extraction began only after the included set was finalized at $n = 173$ studies following full text eligibility assessment. Extraction was conducted in two passes. In the first pass, all fields were populated directly from the full text, tables, and figures. In the second pass, completed after a time gap, a randomly selected subset of included studies was re-evaluated to test internal consistency, and any discrepancies were reconciled against the source paper. High risk fields, such as evaluation metrics, horizon labels, target variable definitions, and the stated XAI method, were additionally checked for unit definitions and normalization bases.

2.7.2. Data items

The following 15 data items were extracted from each included study. These extracted fields are listed in Table 4.

2.8. Quality assessment

2.8.1. Quality assessment checklist

Each included study was assessed using a 10-item quality checklist designed for evaluating XAI applications in energy forecasting. Each criterion is scored as Yes (1 point), Partial (0.5 points), or No (0 points), yielding a maximum score of 10 out of 10. The checklist criteria are presented in Table 5.

Table 4
Extracted data items (D1-D15).

ID	Field	Description
D1	Bibliographic Info	Authors, year, title, journal, DOI
D2	Study Objective	Primary aim and scope of the study
D3	ML/DL Model(s)	Architecture(s), hyperparameter optimization method
D4	XAI Technique(s)	Specific technique(s) applied (e.g., TreeSHAP, KernelSHAP, LIME)
D5	XAI Category	Post-hoc or ante-hoc, local or global, model-agnostic or model-specific
D6	Forecasting Horizon	Ultra-short (<1 h), short (1-6 h), day-ahead, medium, long
D7	Input Features	Meteorological, temporal, historical PV, satellite or image
D8	Dataset Details	Source, size, temporal resolution, location, time span
D9	Performance Metrics	RMSE, MAE, R^2, MAPE, nRMSE, with numeric values and reported units or normalization basis
D10	Key XAI Findings	Top features identified, interaction effects, insights
D11	Validation Method	Train test split, k-fold, walk-forward, hold-out
D12	Deployment Context	Real-time, edge, cloud, proof-of-concept, operational
D13	Limitations	Self-reported limitations and threats to validity
D14	Geographic Region	Country, city, Köppen climate zone
D15	Target Variable Class	Irradiance (GHI, DNI, DHI), solar radiation aggregates, PV power output, or PV energy, including reported unit

Table 5
Quality assessment criteria.

Item	Criterion	Scoring
Q1	Research objectives clearly stated?	Yes = 1, Partial = 0.5, No = 0
Q2	Dataset adequately described (source, size, features, temporal resolution)?	Yes = 1, Partial = 0.5, No = 0
Q3	ML/DL model architecture clearly described and reproducible?	Yes = 1, Partial = 0.5, No = 0
Q4	XAI technique properly explained and justified?	Yes = 1, Partial = 0.5, No = 0
Q5	Appropriate evaluation metrics reported with numeric values?	Yes = 1, Partial = 0.5, No = 0
Q6	Validation methodology sound (no data leakage, appropriate for time-series)?	Yes = 1, Partial = 0.5, No = 0
Q7	XAI analysis provides meaningful insights beyond superficial plots?	Yes = 1, Partial = 0.5, No = 0
Q8	Results compared against baselines or state-of-the-art?	Yes = 1, Partial = 0.5, No = 0
Q9	Limitations and threats to validity discussed?	Yes = 1, Partial = 0.5, No = 0
Q10	Study reproducible (code or data availability stated)?	Yes = 1, Partial = 0.5, No = 0

2.8.2. Quality thresholds

The studies were categorized into three quality tiers based on their aggregate score. High quality studies have scores between 8.0 and 10.0, medium quality studies score between 6.5 and 7.5, and low quality studies score below 6.5. No studies were excluded solely based on quality score, instead quality assessments inform the synthesis in two ways. First, quality scores are reported alongside findings to contextualize the evidence. Second, the main frequency rankings and thematic findings are re-computed within each quality tier, so that readers can verify which conclusions persist when the corpus is restricted to high-quality studies; this stratified analysis is reported in Section 3.1.4. This approach follows established systematic review practice, using quality assessment for evidence weighting rather than exclusion. The resulting quality distribution and per-criterion analysis are reported in Section 3.1.3.

2.9. Data synthesis and analysis methods

Given the heterogeneity of ML architectures, XAI techniques, datasets, and evaluation metrics across the included studies, a formal meta-analysis with pooled effect sizes is not feasible. Instead, synthesis combines three complementary analytical approaches, qualitative narrative synthesis, bibliometric analysis, and taxonomic classification.

2.9.1. Metric harmonization and normalization

Cross study comparability is limited by heterogeneous targets, units, horizons, and denominators. To reduce misleading comparisons, metrics were harmonized using a fixed priority rule. First, results were summarized within target class, irradiance components, solar radiation aggregates, PV power output, and PV energy. Second, whenever a study reported normalized errors, nRMSE, NMAE, or a clearly defined percentage error, those normalized values were used for cross paper summaries within the same target class. Third, when only absolute errors were reported, they were not converted unless the paper explicitly reported a valid normalization denominator.

When a denominator was available, normalization followed a consistent hierarchy. For PV power output, preference was given to rated capacity P_{rated} when stated. If P_{rated} was not provided, the mean target $\bar{y}$ was used only when explicitly reported. Range based normalization using $(y_{\max} - y_{\min})$ was applied only when the paper reported both values. For irradiance and solar radiation targets, normalization used the paper stated baseline, such as clear sky normalization, mean normalization, or range normalization, without substituting external estimates. Studies that did not define their normalization basis were treated as non comparable for cross study metric aggregation, and were only used in narrative synthesis.

The normalization definitions used in this review are

$$\text{nRMSE} = \frac{\text{RMSE}}{d} \times 100, \quad \text{NMAE} = \frac{\text{MAE}}{d} \times 100 \tag{1}$$

where d is the study reported denominator, typically P_{rated}, $\bar{y}$, or $(y_{\max} - y_{\min})$.

2.9.2. Qualitative narrative synthesis

The qualitative narrative synthesis organizes findings thematically around the six research questions (RQ1-RQ6). For RQ1, studies are grouped by model family (LSTM and BiLSTM, CNN and TCN, Transformer, ensemble and tree based, hybrid), and predictive performance is compared using reported metrics. For RQ2, studies are categorized by XAI method (SHAP, LIME, PFI, PDP, Grad-CAM, attention visualization, inherently interpretable), and the quality of explanations is evaluated based on depth of analysis, visualization quality, and actionable insights. For RQ3, XAI-derived feature importance rankings are aggregated across studies to identify consistently important variables. For RQ4, studies are mapped to their temporal scope, and XAI usage patterns are compared across horizons. For RQ5, studies are mapped to countries, regions, and Köppen climate zones. For RQ6, self-reported limitations and evidence-based gaps are synthesized into a structured research roadmap.

2.9.3. Bibliometric analysis

The bibliometric analysis examines publication trends, contributing journals, and keyword co occurrence patterns across the included studies. Annual publication counts are evaluated to trace growth trajectories, and to identify key inflection points. Journal level analysis highlights the principal venues shaping this field. To better capture the research landscape and surface emerging themes, a dual level keyword co occurrence analysis was performed in VOSviewer, separating the broader problem domain from the specific methodological implementations.

Fig. 3 shows the network constructed from standard bibliographic metadata, including titles, abstracts, and author keywords. This macro level view maps the application space well, with large central nodes such as forecasting, deep learning, and attention mechanisms. Yet this conventional abstraction can mask interpretability methods, because authors often favor broad descriptors over naming concrete techniques in abstracts and keyword lists. To address this limitation and better represent the methodological core, a second network, Fig. 4, was generated from the synthesized full text extraction document. This micro level analysis reduces keyword fragmentation, and it also avoids abstract driven generalization.

As visualized in the lower cluster of Fig. 4, SHAP, LIME, feature importance, and XAI appear as dominant and highly connected concepts, which aligns with, and strengthens the quantitative findings of this review. Taken together, the two networks illustrate why full text synthesis is important when estimating the real prevalence of specific Explainable AI algorithms in the literature. Finally, database contribution analysis measures both unique contributions and overlap across the searched databases, supporting practical recommendations for future search strategies in this domain.

2.9.4. Taxonomic classification framework

A novel dual-axis taxonomy is proposed to systematically classify the included studies along two dimensions, the ML model family on the horizontal axis and the XAI technique category on the vertical axis.

The ML model families include

1. LSTM, BiLSTM, and GRU, representing recurrent architectures,
2. CNN and TCN, representing convolutional architectures,
3. Transformer and other attention-based architectures,
4. Ensemble and tree-based models such as Random Forest, XGBoost, LightGBM, and CatBoost,
5. Hybrid or multi-model approaches combining two or more architectures,
6. Other models including SVR, MLP, Bayesian models, and Gaussian Processes.

The XAI technique categories include

1. SHAP and other Shapley-based methods,
2. LIME,
3. Feature Importance and Permutation Feature Importance,
4. Partial Dependence Plots and Individual Conditional Expectation (PDP/ICE),
5. Grad-CAM and saliency-based methods,
6. Attention visualization,
7. Inherently interpretable models,
8. Multiple or combined XAI techniques.

This dual-axis taxonomy enables the identification of well-explored combinations, for example SHAP applied to ensemble models, emerging combinations such as LIME applied to Transformer models, and unexplored gaps including counterfactual explanations for recurrent LSTM-based models.

Each included study is mapped to one or more cells within the taxonomy matrix, providing a structured overview of the research landscape. The taxonomy further reveals whether certain XAI techniques are disproportionately applied to specific model families, which may reflect methodological convenience rather than true technical suitability.

3. Results

This section reports the main outcomes of the systematic review using the finalized corpus of included studies. It first summarizes the PRISMA screening results and the composition of the evidence base. It then presents bibliometric patterns, quality assessment findings, and the taxonomy mapping of XAI techniques against model families. The remaining subsections describe the distribution of ML and DL architectures, XAI methods, input features, forecasting horizons, and performance reporting practices.

3.1. Study selection and corpus overview

This subsection summarizes the screening workflow and the final corpus size used for synthesis.

3.1.1. Study selection

The systematic search across four databases yielded 951 records, Scopus (479), Web of Science (360), IEEE Xplore (72), and ScienceDirect (40). Database contribution and overlap are shown in Fig. 5. After removal of 334 duplicates, corresponding to a 35.1% deduplication rate, 617 unique records underwent title and abstract screening.

Title and abstract screening excluded 399 records. A further 34 uncertain records were reviewed manually and excluded, leaving 184 full text articles assessed for eligibility. Full text review excluded 11 studies under EC6 for superficial XAI implementation, resulting in 173 studies included for qualitative synthesis. Across the included studies, three target classes dominate, irradiance components, PV power output, and time aggregated solar radiation. Results are reported within target class to avoid misleading comparisons across units and physical meanings. The single most frequent exclusion reason was EC2, attention used purely as architecture, accounting for 299 papers, which is 48.5% of the 617 screened records. The study selection workflow is summarized in Fig. 2.

3.1.2. Bibliometric overview

Fig. 7 reveals rapid growth in the field. Between 2018 and 2022, annual publications ranged between 2 and 10 studies per year, followed by a pronounced inflection in 2023 with 19 studies. This upward trajectory intensified in 2024 with 43 publications, and further accelerated in 2025 reaching 73 studies. The 2024–2025 biennium accounts for 116 of the 173 included studies, representing 67% of the total corpus. Key publication venues are summarized in Fig. 9.

In terms of publishers, Elsevier leads with 56 papers, followed by IEEE with 40 papers, MDPI with 32 papers, and Springer Nature with 18 papers, with the remainder spread across other publishers. The publisher distribution is summarized in Fig. 8.

Geographically, 71% of the included studies originated from the top ten contributing countries. China dominated the corpus with 55 studies, accounting for 31.8% of all studies, while India and South Korea ranked second and third with 15 and 10 studies, respectively. Morocco occupied the fourth position with 9 studies, exceeding the USA by two studies, with the USA contributing 7 in total. The distribution of contributions across the top ten countries is shown in Fig. 6.

3.1.3. Quality assessment results

Each of the 173 included studies was evaluated using the 10-item quality checklist described in Section 2.8. Aggregate scores ranged from 5.5 to 10 out of 10, with a mean score of 7.97 and a median of 8.0. Based on the predefined quality thresholds, 114 studies (65.9%) were classified as high quality (score $\geq$ 8), 46 studies (26.6%) as medium quality (score 6.5–7.5), and 13 studies (7.5%) as low quality (score $<$ 6.5). The distribution of quality tiers is presented in Fig. 11.

A per-criterion analysis, shown in Fig. 10, reveals distinct strengths and weaknesses across the corpus. Three criteria achieved universal compliance: Q1 (clear research objectives), Q5 (quantitative evaluation

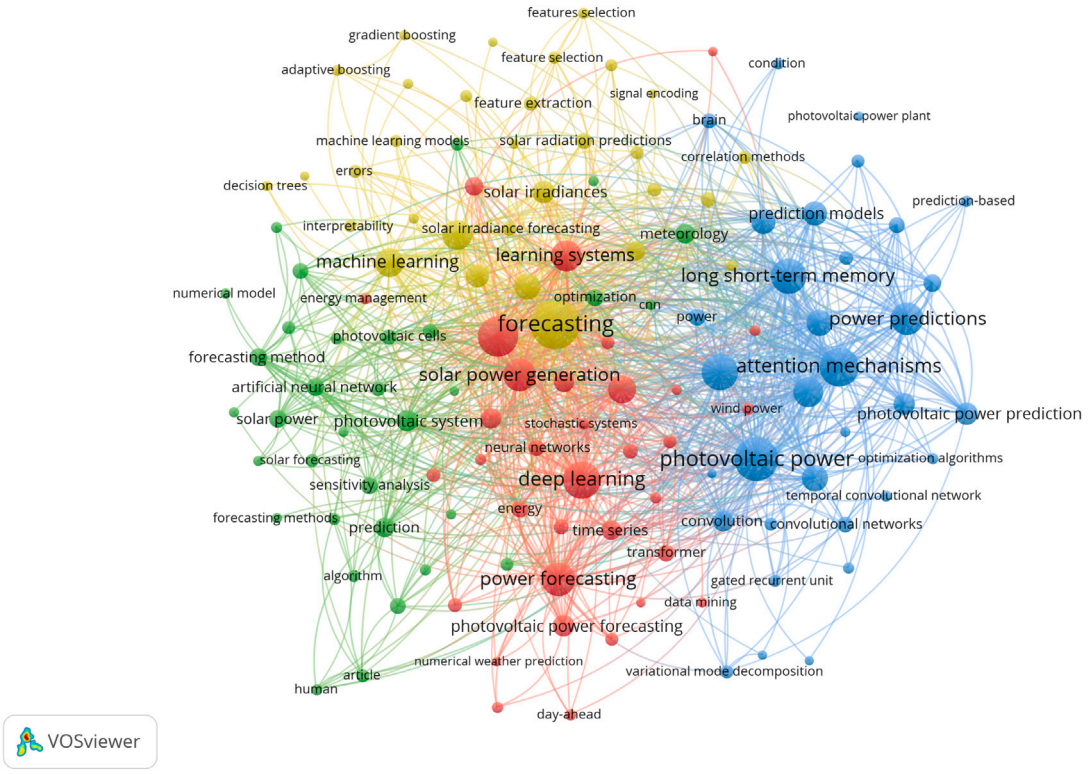


Fig. 3. Macro-level VOSviewer keyword co-occurrence network generated from bibliographic metadata, illustrating the broader application domain and deep learning focus.

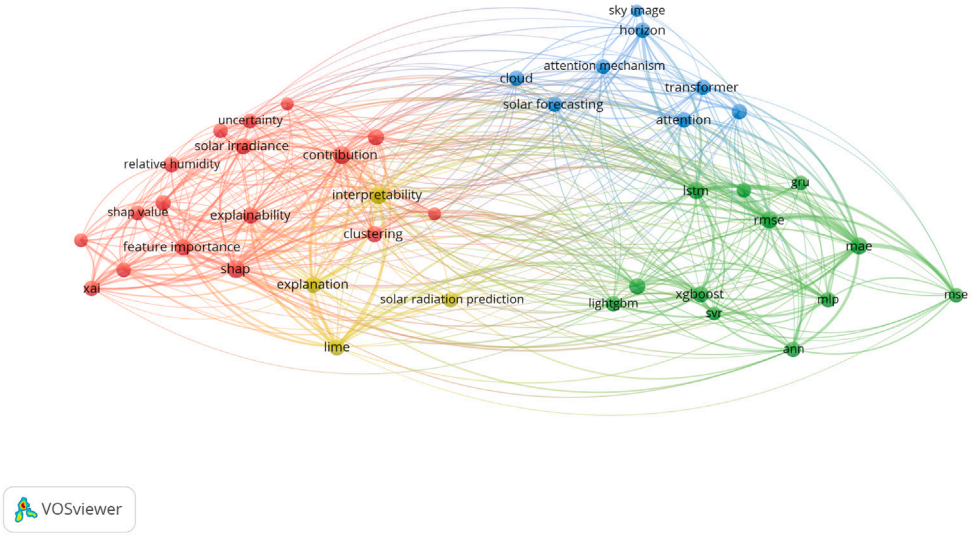


Fig. 4. Micro-level network generated from full-text synthesis, isolating the prevalence of SHAP, LIME, and feature importance within the methodological space.

metrics), and Q8 (baseline comparison), with 100% of studies receiving full scores. High compliance was also observed for Q3 (model reproducibility, 94.8%), Q6 (sound validation methodology, 93.1%), and Q2 (dataset description, 71.1%). In contrast, reproducibility infrastructure remains critically underdeveloped: Q10 (code or data availability) received a full score in only 3 studies (1.7%), with 61.3% scoring zero. Similarly, Q9 (limitations discussion) was fully addressed by only 15.6% of studies, with 65.3% providing only partial discussion. Q4 (XAI technique justification) also revealed a notable gap, with only 57.2% of studies fully justifying their choice of explainability method.

3.1.4. Quality-stratified synthesis

To assess whether the conclusions of this review depend on study quality, the main frequency rankings were re-computed within each quality tier (Fig. 12). The principal findings are robust to quality stratification. SHAP is the most frequently used explainer in every tier: it appears in 53 of 114 high-quality studies (46.5%), 11 of 46 medium-quality studies (23.9%), and 5 of 13 low-quality studies. Notably, the rank order of the five most used technique categories, SHAP, feature importance and PFI, attention visualization, LIME, and intrinsic or rule based models, is identical when computed over the full corpus and when restricted to high-quality studies only. The architecture

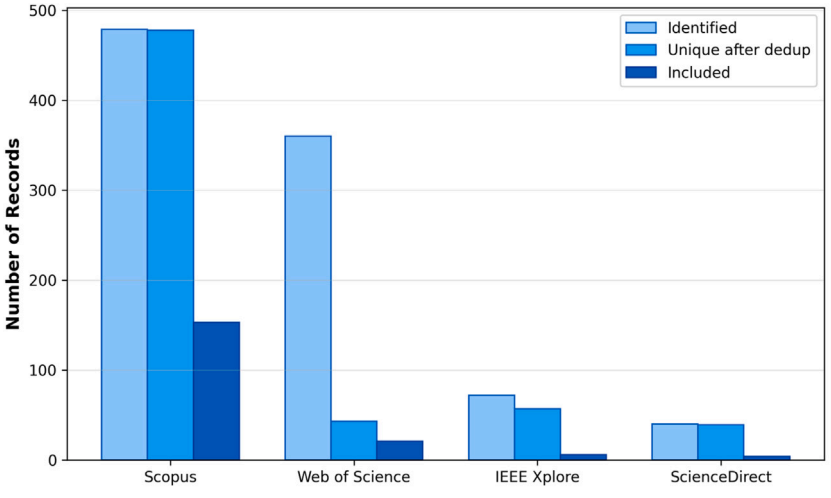


Fig. 5. Database contribution to the systematic review.

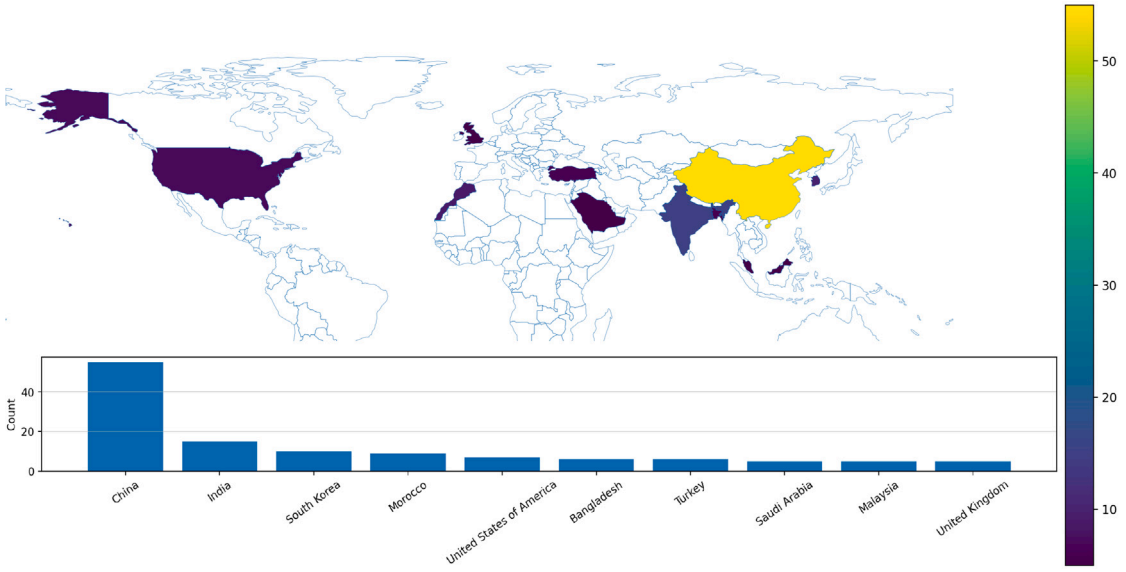


Fig. 6. Geographic distribution of included studies across the top ten contributing countries.

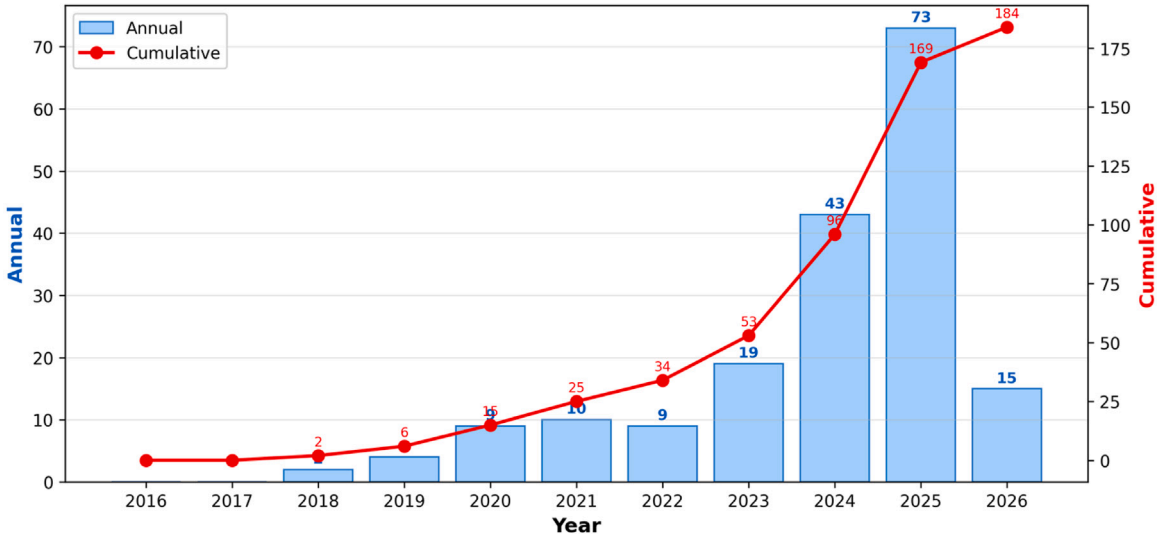


Fig. 7. Annual and cumulative publication trends (2016–2026)

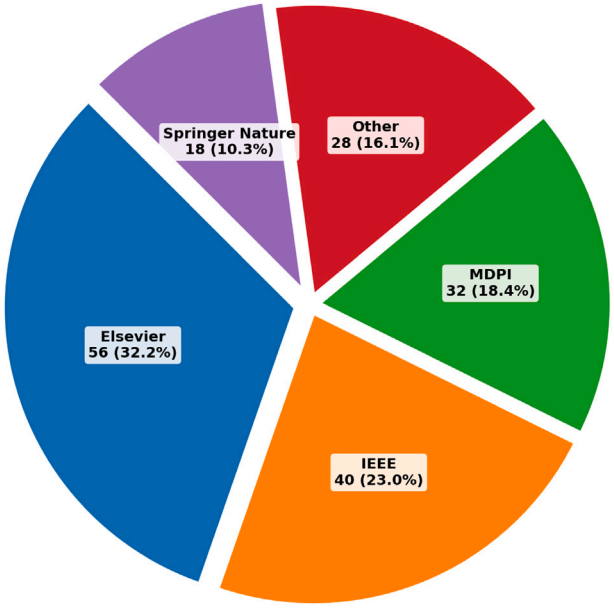


Fig. 8. Publisher distribution across included studies.

landscape is similarly stable: LSTM based recurrent models remain the most frequent family within the high-quality tier (25 studies), and the same broad families, CNN or TCN variants (21), Transformer models (20), Random Forest (19), and feedforward networks (17), populate the next ranks within a narrow band, so no family's prominence depends on lower-quality evidence. None of the thematic conclusions of this review, irradiance primacy in feature attributions, the seasonal and climatic conditionality of secondary drivers, and the scarcity of quantitative explanation evaluation, derives disproportionately from low-quality evidence; low-quality studies constitute only 7.5% of the corpus and no principal claim of this review rests on them alone.

Quality stratification also localizes the corpus weaknesses. Per-criterion means (Fig. 12b) show that low-quality studies score lowest primarily on reproducible model description (Q3, mean 0.31 versus 1.00 in the high tier), sound validation methodology (Q6, 0.27 versus 0.99), and dataset documentation (Q2, 0.50 versus 0.89). Two patterns warrant emphasis. First, code or data availability (Q10) is poor in every tier, with full availability in only 3 of 173 studies, all in the high tier; reproducibility is therefore a field level deficiency rather than a marker of low-quality studies (Section 4.3). Second, methodologically stronger studies do not automatically discuss their own limitations more thoroughly: the mean Q9 score in the low tier (0.73) exceeds that of the high tier (0.51), indicating that aggregate quality scores measure reporting rigor and do not capture how thoroughly authors discuss their own limitations. The cross-study synthesis in Section 3.6 lists the quality-tier composition of the evidence behind each consolidated finding, so that readers can readily identify which findings rest primarily on high-tier evidence and which do not.

3.1.5. Classification of research domains

Based on full-text analysis of 173 papers, four thematic research domains were identified. These domains structure the synthesis across model architectures, XAI techniques, input features with forecasting horizons, and performance reporting patterns.

A dual-axis taxonomy heatmap, shown in Fig. 13, maps XAI techniques against ML model families, highlighting well-explored methodological combinations alongside critical research gaps. Each domain is examined comprehensively in the following sections.

3.2. ML and DL architecture landscape

This section provides a detailed analysis of the machine learning and deep learning architectures employed for solar PV forecasting with explainability across 173 studies, with 162 studies reporting identifiable architectures used for the frequency recount. Fig. 14 presents the distribution of model families. Throughout this section, reported error metrics are quoted as within-study evidence: datasets, targets, units, horizons, and normalization bases differ across papers, so numeric results from different studies are not directly comparable, and no statement in this section should be read as a cross-study performance ranking (see Section 3.5.1).

3.2.1. LSTM and recurrent architectures

Long Short-Term Memory networks are the most frequently used architecture family in this corpus, appearing in 48 studies, reflecting their ability to capture temporal dependencies in solar time series. A representative application is the LSTM-SHAP framework proposed by [70], which used six years (2018–2023) of hourly meteorological data from the Central Weather Administration in Taiwan's Penghu region to predict solar radiation. The framework constructed multivariate input features including temperature, humidity, pressure, wind conditions, and global radiation. SHAP guided feature selection identified 15 key features and reduced the input dimensionality with almost no change in forecasting accuracy, demonstrating that a minimal but well chosen feature set can match the performance of comprehensive inputs.

The study in [71] proposed a novel two-stage hybrid framework coupling a lightweight Transformer-based encoder–decoder (SolarTrans) with generative Large Language Models for both forecasting and interpretability. Trained on 34 days of operational data from two PV plants available on Kaggle, the SolarTrans model achieved MAE of 0.0782, RMSE of 0.1760, and high R^2 on Plant 1, while yielding MAE of 0.1105 and RMSE of 0.3189 on Plant 2. The integration of LLM based natural language explanations alongside SHAP and attention visualizations represents an emerging paradigm where explanations are generated in human readable form for solar farm operators.

For distributed PV plants lacking on-site meteorological data, authors of [72] proposed a GA optimized Bidirectional LSTM (GA-BiLSTM) that exploits spatial correlations from adjacent PV plants instead of weather stations. Applied to PV stations in microgrid settings, the model achieved RMSE of 0.438, 0.806, and 1.118 for 5 min, 15 min, and 30 min ahead predictions respectively, with corresponding R^2 values of 0.983, 0.941, and 0.749. This study reports improved forecasting performance in settings where on-site meteorological data are unavailable by leveraging information from adjacent PV plants.

The CNN-DNN hybrid proposed by [73] used 12,410 data points from 17 geographically diverse regions to achieve RMSE of 74.95 and MAE of 42.87, outperforming standalone XGBoost, GAM, SVR, and CNN or DNN models, with SHAP identifying maximum wind speed, average relative humidity, and highest temperature as key meteorological features.

The interpretable LWDNN model [74] applied to a grid connected PV system in Dhaka achieved RMSE of 6.180 kW, MAE of 1.939 kW, and R^2 of 0.988 using SHAP based feature selection. Notably, a LIME based variant (M3) achieved slightly different results (RMSE = 5.6447 kW, R^2 = 0.990), illustrating how different XAI techniques can guide feature selection toward marginally different model configurations.

3.2.2. Convolutional neural networks

CNNs and its variants appear in 32 studies, employed both for structured meteorological time series (1D CNN) and unstructured sky or satellite imagery (2D and 3D CNN). The 1D CNN framework by [75] was designed for day ahead solar irradiance prediction using data from the Global Ensemble Forecast System and the KENT Mesonet station as part of the AMS Solar Energy Prediction Contest. Enhanced with

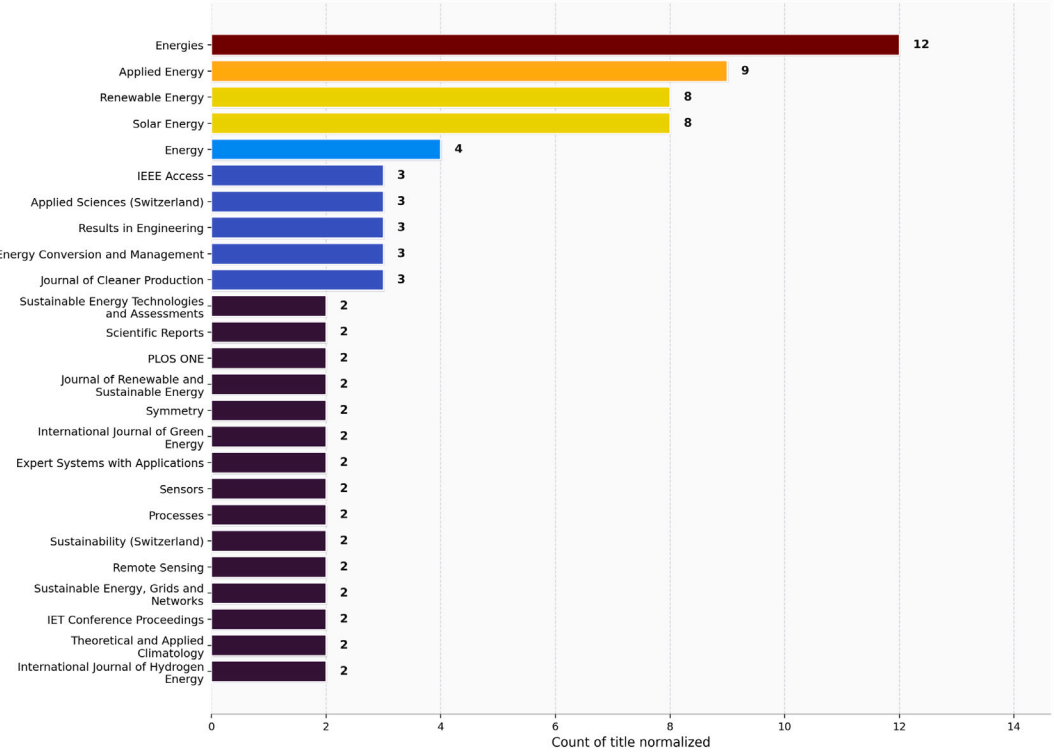


Fig. 9. Top journals by number of included studies.

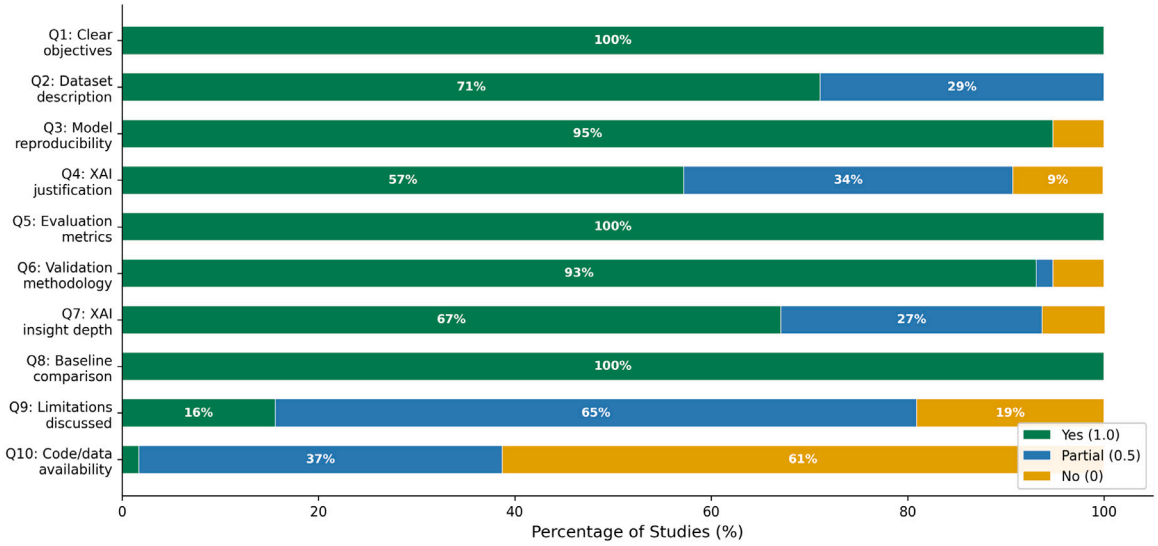


Fig. 10. Per-criterion quality assessment scores across 173 included studies.

residual inception squeeze and excitation blocks and attention mechanisms, the model reduced MAE by 6.56% over the contest best model. Integrated Gradients and Permutation Feature Importance identified solar irradiance and humidity as the key predictive features.

For sky image based ultra short term forecasting, the contrastive self supervised learning approach CL-SUNSET [76] was evaluated on two datasets, a Taipei based campus dataset and the Stanford University open source dataset. On the Taipei dataset, CL-SUNSET achieved RMSE of 14.266 W versus the baseline CNN's 17.226 W (17.1% reduction). On Stanford data, it attained RMSE of 3.081 kW with a forecast skill of +5.17%, substantially outperforming the baseline CNN (−14.51% skill). The contrastive pre training phase learned robust feature representations from unlabeled sky image sequences, reducing the need for large labeled datasets.

A multimodal approach by [26] combined a custom ResNet for sky images with meteorological data from NREL, achieving a forecast skill score of 32.84% and RMSE of 54.36 W/m^2 for 10 min ahead GHI prediction. However, accuracy degraded at longer lead times, with RMSE of 76.45 W/m^2 at the 40 min horizon.

The PV nowcasting model by [77] using Stanford University's Sky Image dataset achieved RMSE of 0.461 and R^2 of 0.996 on sunny days, but RMSE of 3.158 and R^2 of 0.824 under cloudy conditions, a common pattern reflecting the challenge of cloud induced variability.

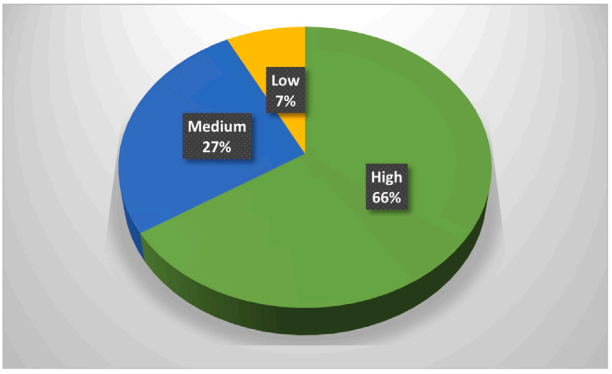


Fig. 11. Distribution of quality assessment tiers across 173 included studies.

The spatiotemporal relation fused network by [60] applied to DKASC data achieved RMSE of 2.46 and MAPE of 1.51% with normalized RMSE as low as 0.61%, while the CNN LSTM model by [78] integrated all sky images with GHI sequences for 10 min ahead predictions.

To address the black box constraint without adding computationally heavy attention mechanisms, [79] proposed a preprocessing methodology that converts 1D time series weather data from Tokyo into 2D images using the Gramian Angular Field transformation. These generated images are then fed into a standard 2D CNN feature extractor (ResNet 18). This approach yields native ante hoc visual interpretability directly through the CNN's convolutional feature maps, visually confirming that the network prioritizes specific physical features like irradiance time series while discarding irrelevant noise, with only a 6% reduction in accuracy compared to a completely opaque baseline LSTM.

3.2.3. Transformer based architectures

Transformer based models appear in 39 studies, reflecting the broader shift toward attention based architectures. Temporal Fusion Transformer is repeatedly reported in this family, and its built in variable selection network can provide feature importance rankings, partially addressing interpretability without relying exclusively on post hoc XAI.

The 2D Temporal Fusion Model by [80] addressed the multi scale nature of solar irradiance by transforming 1D temporal dynamics into a 2D representation. Trained on NREL Solar Radiation Research Laboratory data, it reduced RMSE by 13.8% and MAPE by 21.8% compared to Seq2Seq LSTM for hourly forecasts, and RMSE by 7.6% and MAPE by 21.1% for minutely forecasts. The model provided interpretability through sensitivity analysis and SHAP based rule extraction.

The graph enhanced Transformer by [81] used NREL's Solar Power Data for Integration Studies to model spatial correlations among multiple PV nodes via graph neural networks, outperforming state of the art methods in multi site forecasting scenarios with RMSE of 0.269 at short horizons.

The FNO BiLSTM NAM hybrid [82] applied to data from a Loanda PV Farm (Brazil) achieved MAE of 0.0712 and MSE of 0.0092, incorporating a Neural Additive Model component for inherent interpretability.

The nationwide Korean meteorological station dataset was used by [83] with a TFT based approach, where the model estimated periodic components across all stations for stable prediction. The lightweight Chebyshev polynomial architecture by [84] achieved R^2 of 0.996 and reported edge deployable efficiency (3.90 GFLOPs), with Chebyshev weight visualization providing quantitative interpretation of variable contributions.

3.2.4. Ensemble and tree based models

Tree based ensemble models, Random Forest (36 studies), XGBoost (27), Gradient Boosting (17), LightGBM (12), and CatBoost (12), are favored for their natural compatibility with TreeSHAP, which computes exact Shapley values in polynomial time, eliminating approximation errors that can arise in model agnostic explainers applied to deep learning systems.

A comprehensive comparison by [31] using publicly available hourly PV generation data (April 2012 to July 2014) found that LightGBM achieved the lowest RMSE of 0.08803, MAE of 0.052896, and R^2 of 0.842645, outperforming XGBoost (RMSE = 0.092295, R^2 = 0.827028) and Random Forest (RMSE = 0.091821, R^2 = 0.829). Importantly, this study applied five XAI methods (SHAP, LIME, PDP, Grad CAM, and PFI), finding that while PFI identified the key features, SHAP showed how each feature affected individual predictions, providing complementary insights.

For daily global solar radiation estimation in Fez, Morocco, [85] demonstrated that LightGBM (R^2 = 0.9377, RMSE = 0.4827 kWh/m^2) provided comparable accuracy to SVR (R^2 = 0.9370, RMSE = 0.4855 kWh/m^2) while being inherently more interpretable. PFI and SHAP revealed that extraterrestrial radiation (H_0) and sunshine duration fraction were the two most important parameters.

Researchers in [32] used ensemble boosting techniques on data from 32 meteorological stations, where feature importance analysis identified sunshine hours as the most influential variable, with LightGBM and CatBoost achieving the highest R^2 values.

The XGBoost model by [74] at two PV parks in Southern Germany used SHAP for probabilistic day ahead forecasting, revealing complex nonlinear interaction effects between irradiance and temperature that linear models fundamentally cannot capture.

For solar irradiance forecasting in South Korea, [86] compared XGBoost and LightGBM on Gwangju data (2011–2020), finding that XGBoost achieved MAE of 0.288, RMSE of 0.390, and R^2 of 0.837. SHAP analysis identified hour of day, humidity, and temperature as the most influential features.

The gradient boost ensemble for Indian climate zones by [80] achieved R^2 = 99.38% and identified key meteorological parameters across diverse climatic conditions using feature importance analysis.

In Algeria, the study in [87] assessed the influence of feature selection methods (Pearson, Spearman, mutual information, LASSO, SHAP) on GHI forecasting across two contrasting climates, finding that SHAP based selection yielded the lowest errors in Ghardaïa (nMAE = 17.17%) at the 6 h horizon.

3.2.5. Emerging architectures

Physics Informed Neural Networks appear in 4 studies, embedding physical laws directly into the model architecture and providing intrinsic interpretability [88]. Furthermore, deep reinforcement learning frameworks have recently been adapted to provide interpretable predictive control [89]. However, standard PINNs often rely on fixed physical constraints that fail when weather conditions abruptly change. To resolve this, the Dynamic Parameter PINN proposed by [90] actively adapts the embedded physics such as photoelectric efficiency and temperature coefficients depending on the clustered weather regime, reducing RMSE by 50.8% under extreme weather compared to standard CNN LSTM baselines.

The Physics Informed Mamba Network [91] similarly integrated time frequency decomposition and physics constraints, reducing MAE by 38.16% over baseline models.

Alongside physics constraints, direct structural transparency is emerging, the direct explainable neural network by [84] used solar irradiance data from Lyon, France, utilizing a single nonlinear layer to mathematically extract and visualize mapping correlations, achieving CV RMSE between 0.14 and 0.62.

Another notable development in intrinsically interpretable architectures is the application of Kolmogorov Arnold Networks [13]. Unlike

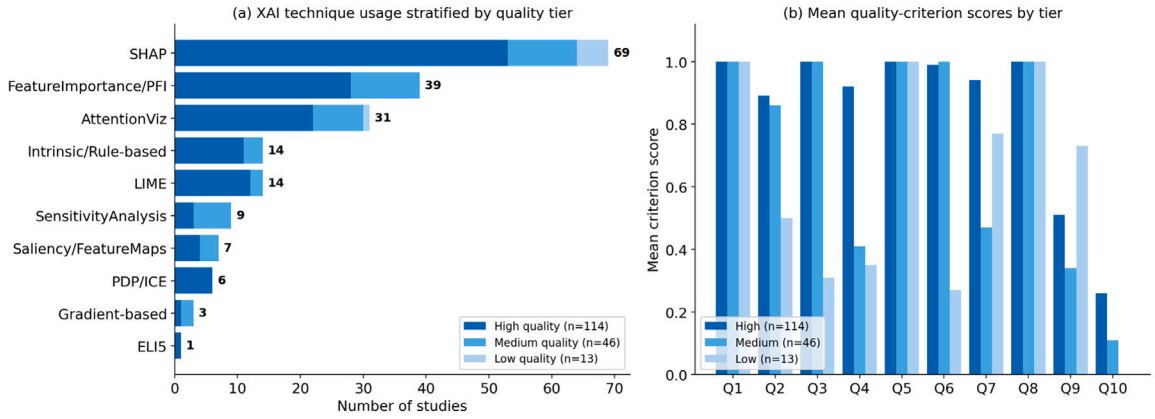


Fig. 12. Quality-stratified synthesis across the 173 included studies. (a) XAI technique usage by quality tier. (b) Mean scores per quality criterion (Q1–Q10) by tier.

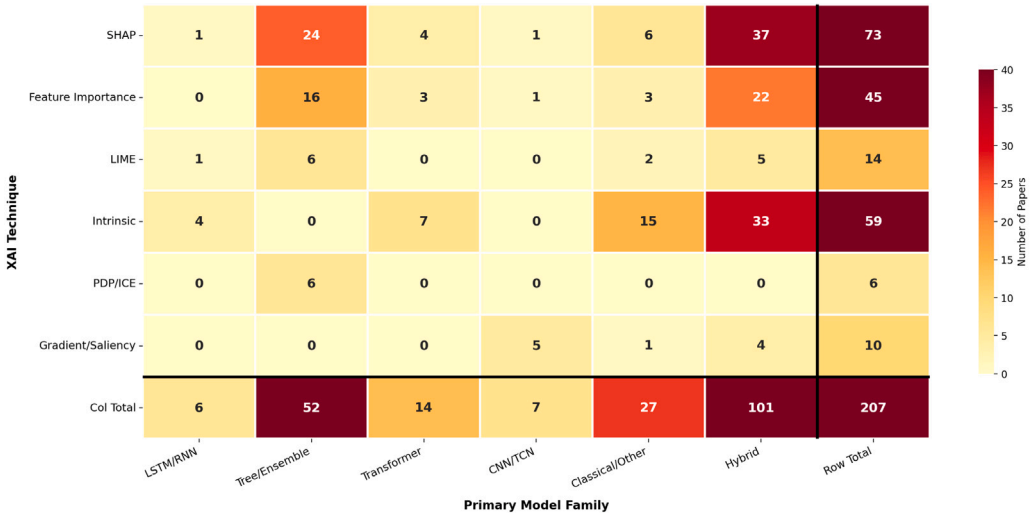


Fig. 13. Heatmap of XAI techniques versus ML model families.

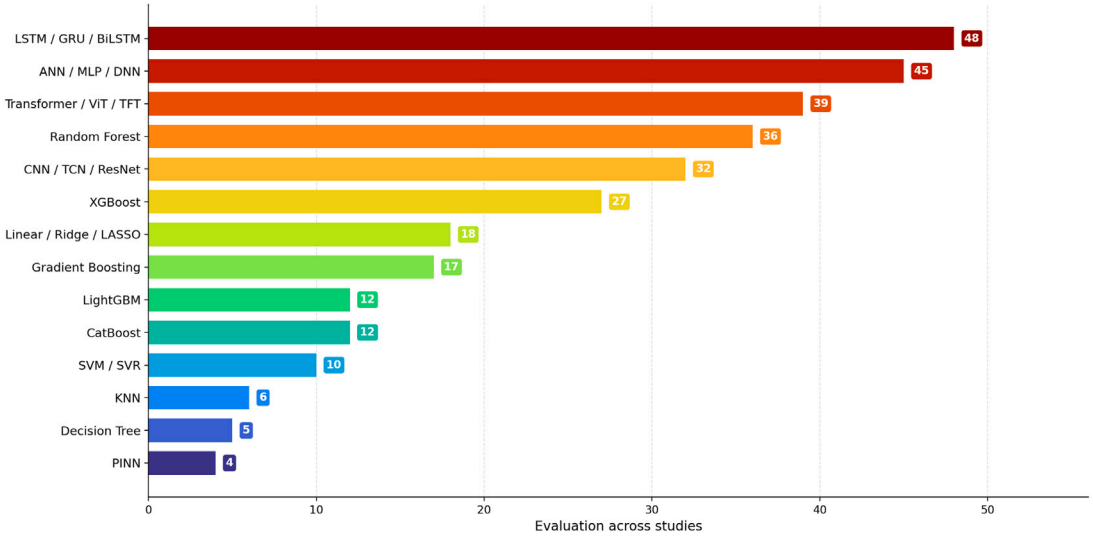


Fig. 14. Distribution of ML/DL architectures.

standard multi layer perceptrons, KANs provide step by step quantifiable computations, maintaining strong predictive accuracy for solar forecasting even when scaled down to a single hidden neuron.

Generative AI is also being utilized beyond basic data augmentation. A novel multi modal framework [69] utilizes a GPT agent not to generate text, but to translate qualitative linguistic weather descriptions such as scattered clouds into numeric likelihoods of cloud cover. These likelihoods are then concatenated with standard numeric features for probabilistic XGBoost forecasting, providing a direct mathematical bridge between natural language meteorological reports and predictive features.

Transfer learning has emerged as a practical approach for newly installed PV plants with limited data. The work done in [92] proposed training a deep model on large auxiliary datasets and fine tuning with a transfer learning technique for a new 400 kWp plant in the Himalayan region of India.

The XCFL framework [93] incorporated SHAP based feature level contributions into federated cluster aggregation, improving model accuracy for heterogeneous PV sites without sharing raw data.

Finally, GAN augmented approaches like [94] used enhanced TimeGAN with WGAN GP and CGAN for PV data augmentation and imputation, with SHAP analyzing latent space representations.

3.2.6. Feedforward Neural Networks (ANN, MLP, DNN)

Standard Artificial Neural Networks and Multi Layer Perceptrons remain highly prevalent, appearing in 45 studies, serving as both robust standalone predictors for tabular data and standard baselines. For macro scale feasibility planning, [95] demonstrated that a classic ANN, when fed with comprehensive meteorological and geographic inputs (notably site elevation, which contributed 14.66% of the predictive variance), can produce a credible monthly average daily solar radiation estimator across diverse climates in Saudi Arabia.

To streamline these dense networks, a study published by authors of [96] applied both LIME and SHAP to explain a trained ANN, using the resulting feature importance rankings to prune low value inputs like certain uncertainty variables. Retraining the optimized ANN on the reduced feature set preserved near perfect accuracy ($R^2 = 0.9980$) while the cited study reports reduced inference overhead and operation count.

Furthermore, the work done in [60] utilized a Feedforward Neural Network paired with LIME to explicitly assess the performance ratio of grid connected PV systems, transparently visualizing the exact influence of environmental degradation factors, such as dust accumulation and soiling loss, on system reliability.

3.2.7. Classical, statistical, and Fuzzy logic models

While deep learning dominates, several studies explicitly reject opaque architectures in favor of highly interpretable, lightweight mathematical systems. For compute constrained streaming contexts, [8] demonstrated that a classical statistical pipeline combining Hierarchical Factor Analysis and Ridge Regression can compress high dimensional meteorological data into three latent constructs, irradiance, temperature, and wind or climate factors. This approach yields lightweight inference and stable, interpretable additive contributions that rival tree ensembles.

For macro scale PV planning where data is highly limited and uncertain, conventional data hungry learning is inappropriate. Here, an optimized nonlinear grey Bernoulli model [97] applied a damping accumulated generating operation to smooth raw sequences, reporting substantial error reductions relative to its baselines without requiring large datasets.

Finally, to bypass neural weights entirely, [98] proposed a Double Input Rule Modules Stacked Deep Fuzzy Model. By relying exclusively on highly interpretable fuzzy if then rules constructed via the least square method, the architecture preserves logical transparency, allowing human experts to directly read and comprehend the rule base governing the PV power predictions.

3.3. Analysis of XAI techniques

Fig. 15 shows the distribution of XAI techniques across 173 studies. This section provides a detailed analysis of how each technique has been applied, what findings it has produced, and its strengths and limitations in the solar forecasting context.

A structured synthesis of model families, operational challenges, and methodological innovations is provided in Table 6.

3.3.1. SHAP

SHAP is the most frequently used XAI technique in this corpus, appearing in 69 studies, about 40% of the 173 included studies. Beyond post hoc explanation, several studies have used SHAP as an active component in the forecasting pipeline. The SHAP EMS framework by [59] directly embedded SHAP interpretability into real time energy management for residential solar battery systems. A hybrid LR XGBoost model provided one hour ahead PV forecasts, while a SHAP weighted rule based controller dynamically adjusted decision priorities based on feature importance. The framework achieved 18.3% reduction in peak grid demand by adapting to temperature fluctuations and generation variability.

SHAP guided feature selection was used by [70] in Taiwan to prune the LSTM input feature set to a minimal subset without accuracy loss. The seasonal variability study by [99] on NREL NSRDB data (24.25°N, 45.34°E) showed that while GHI, DNI, and DHI consistently dominated predictions (correlation > 0.95, SHAP $> 10^{-1}$), secondary features exhibited pronounced seasonal shifts, humidity was critical in summer but less so in winter.

For multi horizon forecasting, the interpretable hybrid by [101] combined CNN LSTM with TFT architecture, where SHAP analysis revealed that the trio of hybrids achieved the lowest MAE (approximately 0.551) through complementary feature capture.

In probabilistic forecasting, [74] applied SHAP to NGBoost at two PV parks in Southern Germany, uncovering complex nonlinear interaction effects between irradiance, temperature, and other variables. Another study in [102] used SHAP to analyze a LASSO RFR hybrid model, finding temperature to be the second most important predictor after solar irradiance, with RMSE of 23.7, 18.2, and 20.8 Wh/m^2 for thin film, monocrystalline, and polycrystalline PV technologies respectively. For urban PV generation, [103] used SHAP analysis on a two stage ML framework with climatic data from four Korean urban buildings, identifying direct solar irradiance, diffuse irradiance, hour of day, visibility, and relative humidity as common key drivers.

Beyond standard feature attribution, SHAP is being pioneered for entirely new operational and security applications. A critical limitation of SHAP in operations is its high runtime overhead reported in some settings, to resolve this, [42] utilized K Means clustering to compress the dataset into representative centroids before applying SHAP, and the study reports a reduction in explanation runtime from 870 min to seconds, without a reported loss in explanation fidelity. Furthermore, authors of [95,104,105] utilized SHAP not for human interpretation, but as an active cybersecurity defense mechanism against data poisoning in smart grids. By using SHAP to extract Spectral and Shannon Entropy feature signatures, the framework successfully filtered out malicious data perturbations that standard anomaly detectors missed. Finally, the work performed in [106] advanced the mathematical foundation of SHAP by introducing the A PDD SHAP algorithm for Asymmetric Shapley Values. By injecting partial causal knowledge (causal DAGs) into the explanation, this technique mathematically forces the model to attribute importance to root physical causes rather than merely correlated proxies, effectively bridging the gap between correlation based XAI and causal inference. Operational roles and typical deployments for each XAI technique are summarized in Table 7.

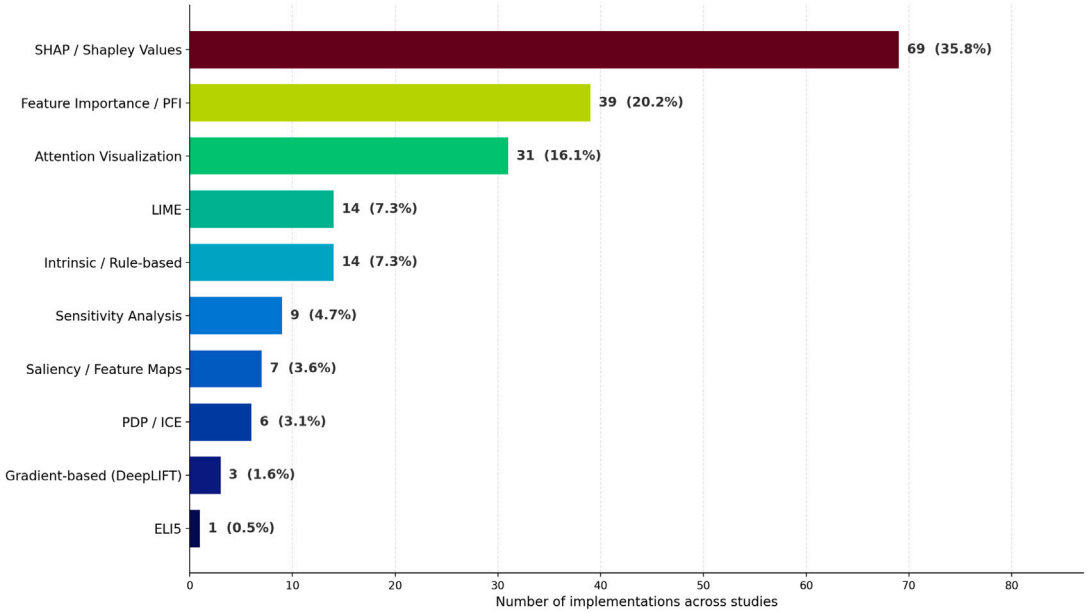


Fig. 15. The distribution of XAI techniques across studies.

Table 6
ML/DL families, target challenges, methodological innovations, and associated studies.

ML model family	Target forecasting challenge & Operational context	Key methodological innovations & Findings	Associated studies
LSTM & Recurrent Networks (BiLSTM, GRU)	Modeling complex, long range temporal dependencies in structured meteorological time series.	[70] uses SHAP guided pruning to reduce the input set to a minimal subset without accuracy loss. [72] GA BiLSTM exploits adjacent PV plant spatial correlations when local weather data is missing. [71] couples a Transformer based encoder decoder (SolarTrans) with an LLM for natural language interpretation. [74] employs an interpretable LWDNN model in Dhaka, achieving R^2 of 0.988.	[70–74,78,80,82,99]
CNN & Image Based (1D, 2D, 3D CNN, ResNet)	Extracting spatial features from all sky imagery and 1D sequential weather data for ultra short horizons.	[76] CL SUNSET applies contrastive self supervised learning to unlabeled sky images, outperforming baseline CNNs. [79] converts 1D time series into 2D images via Gramian Angular Field for native ante hoc visual interpretability. [75] proposes a 1D CNN enhanced with residual inception squeeze and excitation blocks for the AMS contest. [26] MULSKIN framework combines ResNet sky images with NREL data for 10 min horizons.	[26,60,75–79]
Transformer & Attention (TFT, Graph GNN)	Multi scale temporal modeling and multi site spatial correlation with built in variable selection.	[80] 2D TFM transforms 1D temporal dynamics into a 2D representation to capture multi scale variations. [81] graph enhanced Transformer (G GraphTST) models multi site spatial correlations via GNNs. The study done in [84] Chebyshev polynomial Transformer reports edge deployable efficiency (3.90 GFLOPs). [83] estimates periodic components across nationwide Korean stations.	[71,80,81,83,84]
Tree Ensembles (Random Forest, XGBoost, LightGBM)	High speed processing of tabular data and precise probabilistic forecasting using TreeSHAP.	[31] demonstrates LightGBM outperforming Random Forest and XGBoost in day ahead tracking. [74] uses NGBoost to reveal complex nonlinear interaction effects between temperature and irradiance. The study done in [87] evaluates SHAP versus LASSO feature selection across distinct Algerian climates. [32,80,86] report ensemble models performing best within their respective studies in Bangladesh, India, and South Korea.	[15,31,32,69,74,76, 80,85–87,100]
Intrinsic, Physics Informed & Emerging	Bypassing black box models by embedding thermodynamic laws or transparent mathematical structures.	[90] Dynamic Parameter PINN adapts physical bounds based on shifting weather regimes. [13] Kolmogorov Arnold Networks provide quantifiable computations scaling to a single hidden neuron. [69] XGBoost GPT translates linguistic weather text into numeric cloud likelihoods. [98] DIRM DFM bypasses neural weights entirely, relying on readable fuzzy logic rules.	[8,13,41,60,69,84, 85,90–98]

Table 7
XAI techniques, operational roles, methodological applications, and associated studies.

XAI technique	Operational role & Deployment context	Specific insights & Methodological applications	Associated studies
SHAP (Shapley Additive exPlanations)	Unified global feature ranking, local attribution, and active operational control.	**Active Control:** [59] SHAP EMS framework feeds SHAP weights into a rule based controller to dynamically dispatch residential batteries. **Runtime optimization** [42] reports K Means clustering reducing SHAP runtime from 870 minutes to seconds. **Cyber Security:** [95] extracts Shannon Entropy feature signatures to filter malicious data poisoning. **Causal Modeling:** [106] A PDD SHAP injects causal DAGs to map root physical causes instead of correlated proxies. **Feature Discovery:** [99] identifies winter specific shifts in wind direction influence, [102] ranks temperature as secondary driver for thin film PVs.	[31,42,59,70,71,74,74,85–87,95,99,101–103,106]
Feature Importance & PFI	High efficiency dimensionality reduction, preprocessing, and physics validation.	**Physics Validation:** [76] confirms the interaction of GHI and DNI provides cloud radiation physics cues to XGBoost. **Dimensionality Reduction:** [107] Weighted RFE strips noisy inputs, lowering RMSE significantly. **Data Imputation:** [108] uses RFE to rank temporal variables before Stacked Ensemble Imputation. **CNN Validation:** [75] applies PFI to 1D CNNs to ensure convolutions are driven by downward shortwave flux rather than noise.	[15,31,32,75,76,80,85,107–109]
LIME (Local Interpretable Model Agnostic)	Localized troubleshooting, drift detection, and single instance anomaly diagnosis.	**Degradation Diagnosis:** [60] combines with FFNN to isolate predictive drops caused by local dust accumulation and soiling. **Visual Diagnosis:** [77] explains ultra short term power drops by isolating local cloud boundaries in sky images. **Geographic Profiling:** [38] isolates shifting weather influences across coastal versus inland Australian zones. **Sequence Mapping:** [110] applies LIME to RNNs to assign local weights to forecast spikes.	[9,31,38,60,71,74,77,86,91,96,110–112]
Multi XAI & Emerging (Gradient, Counterfactuals)	Cross validating explanations for maximum trustworthiness and prescriptive planning.	**Trust Verification:** [9] combines six methods (SHAP, LIME, ELI5, PDP, ICE, FI) with Bayesian uncertainty to separate data anomalies from model failures. **Prescriptive Action:** [113] DiCE enables what if voltage scenario testing. **Boundary Mapping:** [63] deploys Integrated Gradients and DeepLIFT to map boundary conditions where PV predictions degrade. **Equation Generation:** [41] MGGP generates transparent mathematical formulas, abandoning post hoc explainers.	[9,26,31,41,60,63,74,77,84,84–86,107,113]

3.3.2. Feature importance and Permutation Feature Importance

Feature importance analysis was employed in 39 studies (approximately 23% of the corpus). While SHAP provides unified additive explanations, traditional feature importance and Permutation Feature Importance remain highly favored for their practical simplicity and directness in evaluating global feature influence.

The Weighted Recursive Feature Elimination framework by [107] used NREL data from Halifax, Canada to demonstrate that optimal feature subsets produced more reliable GHI predictions than using all available inputs, with RMSE values dropping from 2.068 to 1.824 for selected configurations. The interpretable LightGBM model by [85] used PFI alongside SHAP for daily solar radiation in Fez, Morocco (R^2 = 0.9377), finding that extraterrestrial radiation and sunshine duration were the two most important parameters.

Beyond simple ranking, feature importance is increasingly used to validate physical relationships and optimize data pipelines. For instance, study performed in [76] utilized XGBoost's native feature importance scores to mathematically show how the combined interaction of Global Horizontal Irradiance and Direct Normal Irradiance provides the model with critical underlying information regarding cloud radiation physics, supporting the machine learning output in meteorological reality. Similarly, another study [75] applied PFI to a 1D CNN, confirming that downward shortwave radiation flux and surface temperature were the dominant drivers, ensuring the convolutional layers had not overfit to statistical noise.

Several studies embed feature importance directly into the preprocessing phase to construct leaner models. For example researchers of [109] utilized feature importance alongside PCA to calculate and select critical features before training an XGBoost model, effectively reducing overfitting. In dealing with missing data, work done in [108] utilized Recursive Feature Elimination as a foundational explainable mechanism, ranking the mathematical importance of temporal variables before passing them into a Stacked Ensemble Imputation algorithm, thereby simplifying the ensemble pipeline. Another study performed in [15] combined PCA with feature importance plots to provide system operators with a transparent hierarchy of input variables, explicitly showing which daily meteorological parameters drive a given ensemble's predictions.

3.3.3. LIME (Local Interpretable Model-Agnostic Explanations)

LIME was applied in 14 studies, playing a distinctly different operational role than global methods. While global feature importance ranks variables across an entire dataset, multiple studies emphasize LIME's value in generating case specific, localized narratives that act as troubleshooting tools for operators.

For example, authors of [31,45] both coupled LIME with SHAP to provide complementary insights. While SHAP offered a global ranking, LIME clarified local predictions, explaining exactly why a specific day or hour deviated from expectations even when the global ranking would suggest otherwise. This local fidelity is critical for anomaly diagnosis and detecting model drift.

A notable geographic application by [38] evaluated five ML models across three Australian climate zones. CatBoost Nelder Mead with LIME achieved the lowest RMSE in Alice Springs (62.78 W/m^2), while AdaBoost performed best in Darwin (RMSE = 22.93 W/m^2). LIME identified that influential predictors varied sharply by coastal versus inland location. The enhanced Random Forest by [100] used DKASC Alice Springs data and applied LIME to explicitly explain predictions under sudden, localized weather variations.

LIME has also been highly effective in diagnosing specific environmental degradation factors. The study in [60] deployed LIME alongside a Feedforward Neural Network to assess the performance ratio of grid connected PV systems in dust prone environments. By utilizing LIME, researchers could explicitly determine and visualize the exact local influence of dust accumulation and soiling loss on individual predictions. Furthermore, LIME is highly adaptable to sequence models. Researchers of [110] provided a clean template for pairing an RNN with LIME, locally assigning weighted effects to show that surface shortwave irradiance and temperature were impacting specific forecast spikes or dips. Finally, for visual applications, the PV nowcasting model by [77] combined LIME with saliency maps to interpret sky images at Stanford University, isolating the local cloud boundaries impacting immediate power drop offs. Consensus feature categories and their physical roles are summarized in appendix Table A.1.

3.3.4. Other techniques and multi-XAI approaches

Attention visualization (31 studies) represents the third most prevalent XAI category, leveraging the built in attention weights of Transformer and attention augmented architectures to reveal which temporal steps and spatial inputs are prioritized by the model. [26] applied attention visualization alongside SHAP for the multimodal framework, [84] visualized attention patterns in the Chebyshev polynomial Transformer, and TFT based studies such as [41,69] exploited the architecture's native temporal attention heads for inherent interpretability.

Intrinsic and rule based approaches (14 studies) bypass post hoc explanation entirely, embedding interpretability directly into model structure through fuzzy logic rules, Kolmogorov Arnold Networks, Neural Additive Models, and white box statistical models such as MARS.

Sensitivity analysis (9 studies) was used by [84] for the direct explainable neural network, and by [60] for the STRFN model applied to DKASC data, quantifying how perturbations in input variables propagate through model predictions.

Saliency and feature maps (7 studies) visualize CNN activation patterns to reveal which spatial regions of sky images or spectral features drive predictions, as demonstrated by [77] for sky image interpretation and [76] for contrastive learning based feature extraction.

PDP and ICE plots (6 studies) were combined with SHAP and LIME by [86] for comprehensive XGBoost and LightGBM interpretability in Gwangju, South Korea, and by [47] for solar energy prediction using five ML models with training R^2 of 0.608 and testing R^2 of 0.46.

Gradient based methods including Grad CAM, Integrated Gradients, and DeepLIFT (3 studies) were applied in [31] alongside four other XAI techniques, providing pixel level or neuron level attribution maps for deep learning architectures.

ELI5 (1 study) was applied in [6], which paired the ELI5 library with a Random Forest regressor alongside LIME and SHAP.

Of 173 papers, 47 (27%) employed two or more XAI techniques. The most comprehensive multi XAI study [9] applied a six-part explanation pipeline (SHAP, LIME, PDP, ICE, feature importance, and Bayesian uncertainty analysis) to a Bayesian optimized Informer model, achieving MAPE of 6.9% and R^2 of 0.998 after anomaly detection on a Morocco based PV dataset. The study demonstrated that combining model uncertainty with NMAE enables reliable differentiation between data quality issues and model failures. The work performed in study [74] applied multiple techniques (SHAP, LIME, Grad-CAM, and feature importance) to a lightweight DNN for a Dhaka PV system.

Beyond post hoc tools, authors of [63] deployed advanced gradient based methods including Integrated Gradients, Expected Gradients, and DeepLIFT alongside Bayesian uncertainty to explicitly map the boundary conditions where PV predictions fail. Some studies explicitly rejected post hoc explainers in favor of intrinsic white box transparency. For instance the research performed in [41] utilized Multigene Genetic Programming to explicitly generate transparent mathematical equations mapping meteorological inputs to irradiance.

Explainability has also been extended into tail risk modeling. The EVDBM framework [85] embedded XAI directly into extreme value analysis to identify correlation shifts during rare, extreme low production events. Finally, Diverse Counterfactual Explanations appeared in only one study [114] to explain current and voltage predictions, highlighting a critical gap for future research regarding prescriptive what if scenario planning in grid operations.

3.3.5. Critical appraisal of the dominant XAI techniques

The frequency rankings above describe which techniques are used in the literature, not which are reliable. Each dominant technique carries documented limitations that are directly relevant to solar forecasting, and that the included studies rarely acknowledge (Table 8).

Computational complexity. Exact Shapley value computation is exponential in the number of features; the practical estimators differ sharply in cost. TreeSHAP achieves polynomial time exactness for tree ensembles [115], but KernelSHAP on deep sequence models requires extensive model evaluations per explained instance. This cost is documented within the corpus: one included study reported 870 min of SHAP computation on a single dataset before clustering based compression reduced the runtime to seconds [42]. Explanation runtime is reported in only a few of the 173 studies, leaving the feasibility of per-dispatch-interval explanation largely undocumented.

Explanation instability. LIME explanations depend on random perturbation sampling, kernel width, and surrogate complexity, so repeated runs on the same instance can produce different feature rankings [116,117]. Gradient and saliency based attributions for deep networks are fragile to imperceptible input perturbations [118], which is directly relevant to sky-image nowcasting, where sensor noise is commonplace. In this corpus, only three studies quantify attribution stability, for example with Kendall's tau rank correlation across seasons [102,103], so the stability of most published explanations is unknown.

Sensitivity to correlated inputs. Meteorological predictors are strongly collinear, irradiance components correlate with each other, with temperature, and with cloud proxies. Standard SHAP estimators assume feature independence, and under dependence they can assign attribution to correlated proxies or evaluate the model in physically impossible regions of feature space [119]. Permutation importance shares the same defect, because permuting one member of a correlated pair creates unrealistic samples [116]. PDP curves can likewise be distorted when averaging over correlated covariates, and few included studies report the conditional variants (ALE plots, conditional SHAP) that mitigate this. Consequently, the fine-grained ordering of correlated secondary drivers (humidity versus cloud cover versus temperature) reported by individual studies should be read with caution, even though the coarse consensus on irradiance primacy is too consistent across studies to be attributable to these artifacts.

Faithfulness. Post hoc explanations are approximations of model behavior, and can be manipulated: adversarially constructed models can mask their actual feature dependence from LIME and SHAP audits [117]. For attention, the debate is unresolved, attention weights are not automatically faithful importance estimates [66], though they can carry explanatory signal under specific test conditions [67]. This review therefore treats attention visualization as a distinct, weaker evidence class, and the near total absence of fidelity testing in the corpus (Section 3.5.2) means that most published explanations are plausible

Table 8
Documented limitations of the dominant XAI techniques and their implications for solar PV forecasting.

Technique	Main limitations	Implication for PV forecasting	Mitigations/evidence
SHAP (KernelSHAP)	High computational cost; feature independence assumption under correlated inputs [119]; manipulable post hoc [117]	Runtime can prohibit per-interval explanation (870 min reported in [42]); credit may shift among collinear weather variables	TreeSHAP for tree models [115]; conditional/causal variants [106]; clustering compression [42]
LIME	Sampling induced instability; kernel and locality sensitivity [116,117]	Repeated explanations of the same forecast anomaly may disagree, weakening operator trust	Repeat-run stability reporting; cross-check against SHAP [31,74]
PDP/ICE	Misleading under correlated features; hides heterogeneity (PDP) [116]	Average irradiance-response curves can misrepresent regime dependent behavior	ALE plots; ICE alongside PDP [86]
Grad-CAM/saliency	Fragile to small input perturbations [118]; layer choice dependence	Cloud-boundary attributions in sky-image nowcasting may not survive sensor noise	Sanity checks; multi-method visual validation [77]
Attention visualization	Faithfulness contested [66,67]	Attention maps in Transformer forecasters are suggestive, not verified importance	Treat as hypothesis; verify with post hoc attribution [68]
Permutation importance	Unrealistic samples under correlated features [116]	Rankings among collinear meteorological drivers can be distorted	Conditional permutation; grouped features
Intrinsic/white-box	Possible accuracy trade-off; architecture constraints [120]	May sacrifice accuracy under complex cloud regimes	Physics informed hybrids [90,91]; KANs [13]

rather than verified. Where explanation fidelity is critical, inherently interpretable designs remain the more defensible approach [120].

Reproducibility and consistency of explanations. Explanation outputs depend on implementation details that the corpus rarely reports: the SHAP variant and background dataset, LIME sampling parameters, or the layer chosen for Grad-CAM. Different explainers also disagree on the same model and data; consequently, multi-XAI cross-checks [9,31,74] are valuable. Given that only 1.7% of studies release code and data (Section 4.3), most published explanations cannot currently be independently reproduced, even in principle.

3.4. Input feature analysis and forecasting horizons

3.4.1. Consensus feature importance from XAI

A synthesis of XAI derived feature rankings across 173 studies reveals a strong consensus. Solar irradiance (GHI, DNI, DHI) was identified as the most important predictor in nearly all studies, regardless of model architecture, geographic location, or forecasting horizon. This is physically expected as irradiance directly determines PV output power. SHAP analysis by [103] for urban PV found that direct and diffuse solar irradiance were common key drivers alongside hour of day, visibility, and relative humidity. The daily global solar radiation analysis by [85] in Fez, Morocco, confirmed through both PFI and SHAP that extraterrestrial solar radiation and sunshine duration fraction were the overwhelming governing variables.

Temperature emerged as the second most influential environmental variable, reflecting the thermal derating effect on PV cell efficiency. Humidity ranked third, directly affecting atmospheric transmittance. Cloud cover was particularly important for ultra short term forecasting. The minimal feature approach by [121] explicitly challenged the more features assumption. They demonstrated that reliable forecasting can be achieved using only two XAI identified variables, solar irradiance and soil temperature at 10 cm depth, with the latter acting as a proxy for diurnal thermal inertia and environmental state.

A critical finding across the literature is the heavy reliance on temporal and cyclical encodings. In Malaysia, authors of [102] applied SHAP to show that explicit time encoding was the second most influential driver of PV prediction across three different PV technologies, trailing only direct irradiance. Similarly, another study [122] used SHAP to quantify that applying trigonometric cyclical encoding to temporal variables increased the importance of monthly and daily cycle signals by roughly 50%, stabilizing models during high radiation seasons.

Beyond standard numerical features, XAI has validated the use of unconventional proxies. For macro scale feasibility in Saudi Arabia, sensitivity analysis in [95] revealed that site elevation contributed 14.66% of predictive variance, acting as a proxy for atmospheric thickness and temperature gradients. In a notable example, researchers of [123] utilized a modified SHAP framework to validate the translation of human natural language cloud observations into categorical features, showing that textual weather descriptions could boost short term forecasting accuracy by 5% to 15%.

A key finding from [99] is that feature importance rankings shift seasonally. While irradiance components consistently dominate, secondary features like humidity and wind direction show pronounced seasonal variation, with wind direction gaining significant predictive influence specifically during winter. Location dependent rankings were documented by [38] across three Australian climate zones and by [87] across two Algerian climates. In Ghardaïa (arid), MI based feature selection performed best at t+1 (nMAE = 6.44%), while LASSO performed best in Algiers (coastal, nMAE = 10.82%). These findings argue against applying universal feature rankings without location specific XAI validation. The distribution of input feature categories across studies is summarized in Fig. 16. The distribution of evaluation metrics reported in the literature is summarized in Fig. 17.

3.4.2. Forecasting horizons and XAI temporal mapping

Fig. 18 presents the distribution of forecasting horizons across the reviewed literature, while Fig. 19 maps the specific Explainable AI techniques deployed across these distinct temporal scales. To ensure methodological transparency, this cross-tabulation isolates $n = 83$ specific horizon evaluations extracted directly from the subset of studies detailing explicit temporal boundaries. Studies that did not clearly define their temporal horizon in absolute units — including generalized structural analyses utilizing techniques like Partial Dependence Plots (PDP) or Individual Conditional Expectation (ICE) [108,109,113] — were excluded from this specific quantitative mapping to maintain strict rigor and prevent assumption-based classification.

The application of Explainable AI exhibits a strong relationship with the length of the forecasting horizon, as the physical drivers of solar variability shift from transient cloud motion to macro-scale diurnal and seasonal cycles.

Short-Term and Intra-Day Forecasting (1 to < 24 h): Short-term forecasting represents the most densely populated horizon in the literature (41 explicit evaluations). Because models in this regime rely heavily on tabular meteorological inputs — such as temperature, humidity, and explicit time encodings — post-hoc attribution methods are the most common choice. SHAP and Feature Importance are heavily utilized [71,74,124–126] to rank exactly how interacting weather covariates drive daytime generation curves and support grid dispatch.

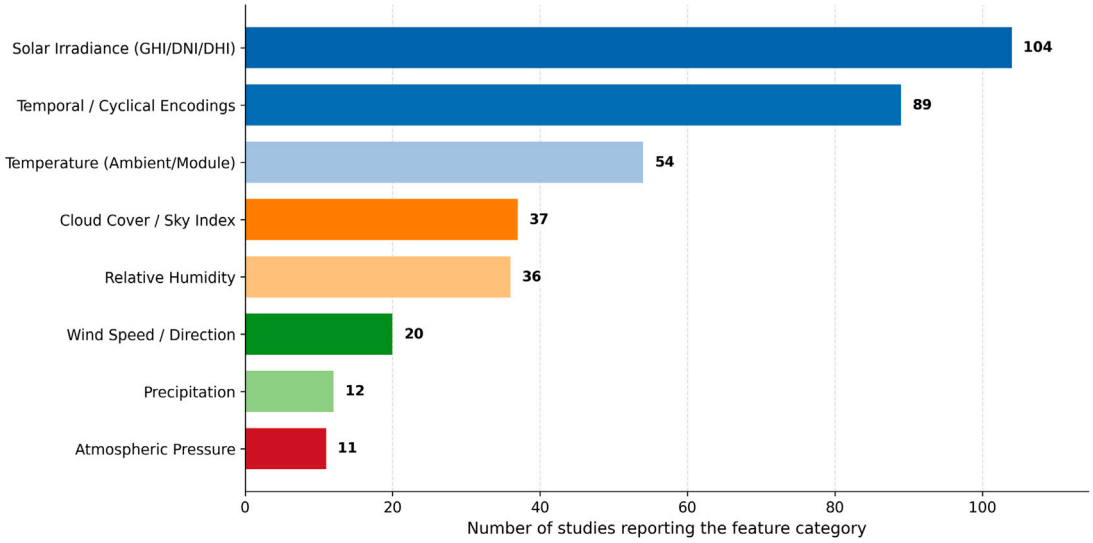


Fig. 16. Distribution of input feature categories across included studies.

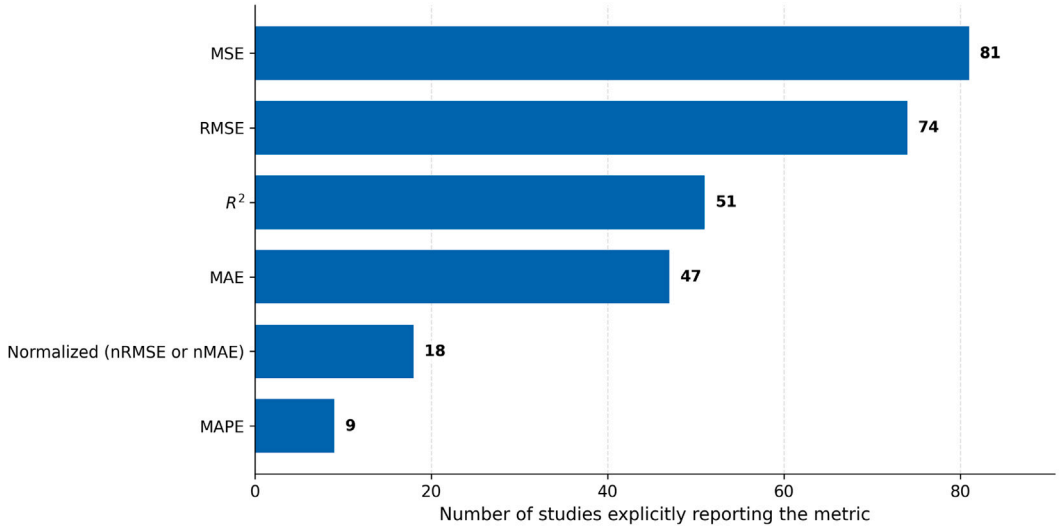


Fig. 17. Distribution of evaluation metrics reported across included studies.

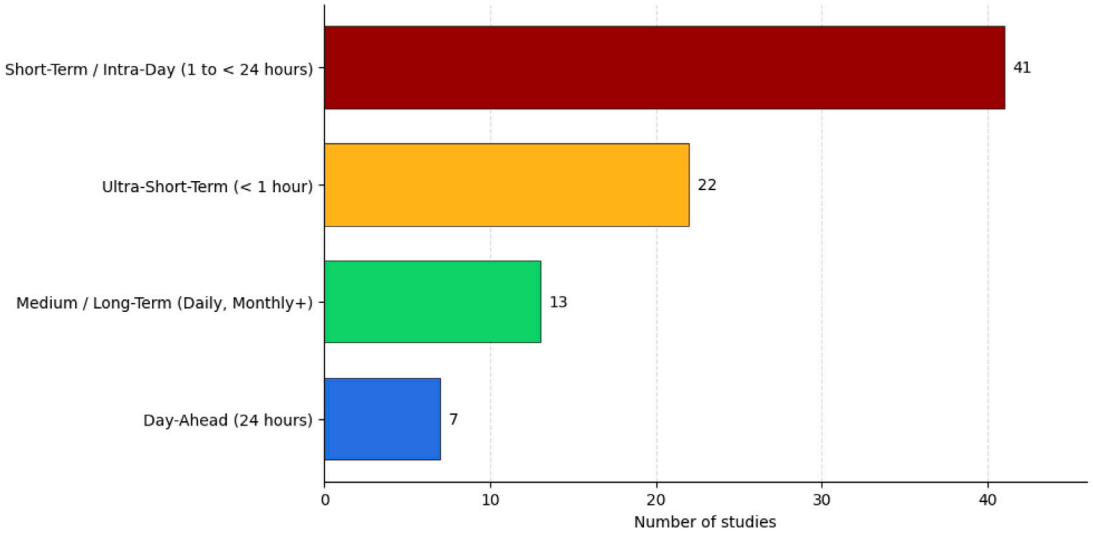


Fig. 18. Forecasting Horizon Distribution Across Studies.

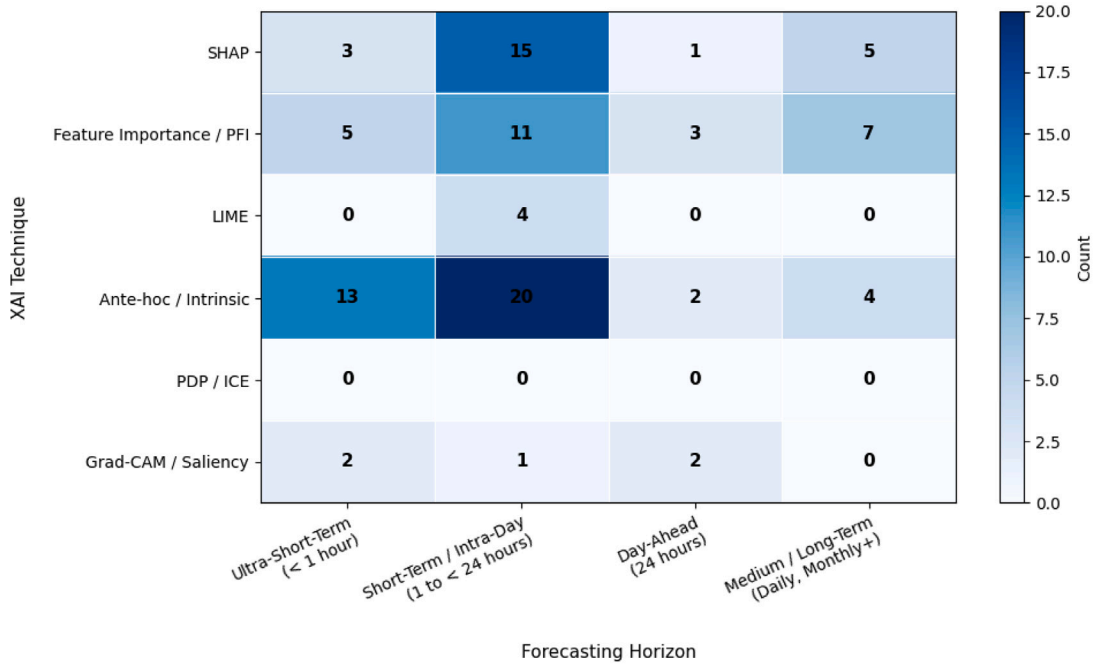


Fig. 19. Dual Axis Analysis of Forecasting Horizon and XAI Techniques.

Furthermore, this is the only horizon where Local Interpretable Model-agnostic Explanations (LIME) see significant deployment [74,125–127], as operators specifically require local troubleshooting to diagnose why a specific hour or day deviated from the expected forecast.

Ultra-Short-Term and Nowcasting (< 1 h): For ultra-short-term horizons, the predictive burden shifts toward capturing rapid, cloud-driven ramps. Here, sky images are critical. For instance, study performed by the authors of [26] achieved a 32.84% forecast skill for 10-minute ahead GHI using sky images, while another study [76] achieved an RMSE of 3.081 kW on Stanford data. Because standard feature attribution struggles to explain spatial image data, XAI application in this horizon relies heavily on Ante-hoc/Intrinsic architectures and visual Grad-CAM/Saliency methods. Researchers utilize embedded attention mechanisms [47,106,128], interpretable Physics-Informed neural networks [91], and visual convolutional mappings [44] to provide evidence that models are tracking true physical cloud boundaries rather than overfitting to pixel noise. The literature clearly demonstrates that predictive confidence and feature utility decay rapidly over time within these short windows. The multi-horizon analysis by [72] showed progressive accuracy degradation, with R^2 dropping from 0.983 at 5 min ahead, to 0.941 at 15 min ahead, and down to 0.749 at the 30-minute ahead horizon. Furthermore, Transformer attention mapping in [68] revealed a dynamic modality shift: at a 2-minute horizon, the model heavily prioritizes the visual spatial features of clouds, but as the horizon extends to 10 min, the model shifts its attention weights away from images and relies almost entirely on numerical clear sky irradiance features.

Day-Ahead Forecasting (24 h): Day-ahead forecasting relies increasingly on Numerical Weather Prediction (NWP) data and satellite imagery [129], while multi-site spatial correlations are heavily utilized to boost accuracy [130]. The 2D-TFM by [80] achieved RMSE reductions of 13.8% over standard LSTM models in this regime. Explainability at the 24-hour mark is primarily deployed to correct structural NWP biases and validate satellite correlations, such as utilizing SHAP to explain day-ahead NWP credibility models [68] or applying Permutation Feature Importance to day-ahead 1D CNNs [75].

Medium and Long-Term Forecasting (Daily, Monthly, Seasonal): At macro scales (13 explicit evaluations), forecasting supports long-term capacity planning, system sizing, and yield projections [131]. Deep learning applications are less common here; instead, XAI is utilized predominantly for structural and physical sensitivity analysis rather than real-time operational troubleshooting. Methods such as SHAP and Permutation Feature Importance are deployed to explain the influence of site elevation on monthly radiation [95], map seasonal correlation shifts between winter and summer regimes [99], and track the impact of daily cyclical encoding over long-term operations [132].

3.5. Performance evaluation and benchmarking

3.5.1. Forecasting metrics

Studies employ heterogeneous metrics, RMSE (138 studies), MAE (127), R^2 (84), MAPE (46), and nRMSE (29). Reported accuracy ranges widely. R^2 varies from 0.46 [47] to 0.999 [108,133]. Absolute errors range from near zero in highly normalized edge cases such as [60] (MAE = 0.0036) to massive unnormalized bounds reported by [73]. These disparities reflect differences in datasets, temporal resolution, system capacity, and preprocessing rather than inherent model superiority.

Critically, reported units vary widely (W, kW, MW, W/m^2) and many studies omit a clear normalization basis, so direct cross study comparison is unreliable. Typical reported metric ranges, organized by target variable class, are summarized in appendix Table A.2; these ranges describe the spread of the literature and must not be read as model rankings. Cross paper comparisons are therefore restricted to normalized errors when the denominator is explicitly defined, otherwise results are summarized within each target class, irradiance, solar radiation aggregates, PV power output, and PV energy, as reported. Few studies report confidence intervals or statistical significance tests, though work done in [28] explicitly utilized p values ($p < 0.001$) to demonstrate that optimization gains were statistically significant rather than noise. Forecasting horizons, metric categories, and supporting evidence are synthesized in Table 9.

3.5.2. XAI evaluation gap

The most significant methodological gap across the literature is the scarcity of quantitative XAI specific evaluation metrics. While 173 studies generated explanations, almost none empirically measured explanation fidelity, stability, or comprehensibility.

Table 9
Forecasting horizons, evaluation metrics, and methodological evidence.

Temporal horizon/Metric category	Application context	Key insights & Methodological evidence	Associated studies
Ultra Short Term (<1 h)	Sky image integration, high frequency grid nowcasting, and contrastive learning.	**Image Models & Nowcasting:** [42,43,134] heavily benchmark sky and satellite imagery for intra-hour forecasting, while [76] proposes contrastive self-supervised learning to process unlabeled sky sequences. **Emerging Architectures:** [91] implements a Physics-Informed Mamba network for ultra-short horizons. [106] introduces multimodal deep learning for very short-term prediction.	[42,43,76,77,91,94, 106,134–136]
Short Term & Day Ahead (1 h to 24 h)	Operational grid dispatch, NWP integration, and probabilistic day-ahead planning.	**Active Dispatch:** [59] applies a 1-hour horizon to dynamically adjust residential battery dispatch based on XAI outputs. **Hybrid & Probabilistic:** [137] utilizes a weather-based attention BiLSTM for day-ahead scheduling. [69] applies interpretable probabilistic forecasting for building-level monitoring. [68] focuses on weather-type credibility for multi-model dynamic combination.	[9,27,59,68,69,86, 126,137–140]
Forecasting Error Metrics (RMSE, MSE, MAE, R^2, NMAE)	Quantifying raw predictive accuracy against baseline models and assessing boundary conditions.	**Normalization inconsistency:** Accuracy bounds vary widely due to unnormalized targets, with R^2 ranging from 0.46 [47] to 0.999 [108,133]. **Advanced Benchmarking:** [85] introduces extreme value analysis and weighted benchmarking to map tail risks. [12] successfully combines NMAE with uncertainty estimation to separate data anomalies from true model failures. [28] is a rare example utilizing p-values ($p < 0.001$) to confirm statistical significance.	[12,15,27,28,60,85, 87,107,108,141, 142]
Quantitative XAI Evaluation Metrics	Validating the mathematical fidelity, stability, and operator trust of explanations.	**Stability:** [102] employs Kendall's tau and Top-k overlap to mathematically prove SHAP stability across different seasons and PV cell types. **Fidelity & Runtime:** [42] explores data clustering to compress inputs, verifying that clustering substantially improves XAI runtime without corrupting explanation fidelity. **Trust Verification:** [13] emphasizes formal methods to enhance trust in ML meteorological predictions, while [71] uses Generative AI (LLMs) to make XAI outputs readable to human operators.	[12,13,71,102,123, 141]

Notable exceptions define the current research frontier. The study done in [103] validated interpretive consistency and transferability of SHAP results across different seasons using Kendall's tau and Top k overlap indicators, while work in [42] explicitly measured explanation fidelity by computing mathematical deviations in SHAP values after dataset compression.

Furthermore, explanation runtime is rarely reported, leaving the real time deployment feasibility of SHAP and LIME unclear for high frequency control. The scholars of [42] documented that standard SHAP computation required 870 min on their dataset, forcing the use of K Means clustering to reduce explanation runtime to seconds for operational viability.

Finally, human in the loop evaluations remain entirely absent. The practical value of explainability is universally assumed rather than empirically tested on actual grid operators.

3.6. Cross-study evidence synthesis

The preceding subsections report evidence study by study. This subsection consolidates it, grading each recurring finding by the number of independent supporting studies, their geographic and climatic spread, and their quality-tier composition (Table 11). Four synthesis questions structure the analysis.

Which architectures are consistently robust? The corpus does not support the claim that any single architecture family is universally superior. Within-study comparisons almost always favor the proposing hybrid, and heterogeneous targets, units, horizons, and normalization bases prevent portable cross-study rankings (Section 3.5.1). What the evidence does support is a conditional pattern that recurs across independent studies: tree ensembles remain competitive and computationally cheap for tabular meteorological inputs [31,32,85,86], LSTM based recurrent models are the most common choice for sequence structure, and sky-image driven CNN variants are consistently the strongest choice at sub-hour horizons [26,76,77], where every study that stratifies by sky condition also reports accuracy degradation under broken cloud regimes [26,77]. Robustness oriented reporting, error stratified by weather regime or season, appears in only a minority of studies, so architecture robustness in the strict sense remains an open evidence gap.

Which XAI techniques are most reliable and informative? Reliability evidence exists at two levels. At the theoretical level, TreeSHAP paired with tree ensembles is the only widely used combination whose attributions are exact rather than approximated [115], which accounts for its frequent use in tabular PV forecasting. At the empirical level, only three included studies quantitatively verify their own explanations, using rank stability indicators such as Kendall's tau across seasons [102,103] or fidelity checks after dataset compression [42]. Informative value is most consistently reported for multi-method pipelines, where a global ranker (SHAP or PFI) is cross-checked by a local diagnostic (LIME or ICE) [9,31,74]; agreement between independent explainers is currently the strongest practical reliability signal available in this literature. Attention visualization, by contrast, should be treated as suggestive rather than reliable, given the unresolved faithfulness debate (Section 3.3.5).

Which approaches perform better under different climatic conditions? Three independent multi-climate studies converge on the

Table 10
Synthesized answers to the primary research questions.

RQ	Answer based on this review	Remarks and Caveats
RQ1	LSTM-based architectures are the most frequently utilized for XAI-linked solar forecasting, with feedforward baselines, rapidly rising Transformer models, tree ensembles, and CNN variants all strongly represented. Tree ensembles (e.g., XGBoost, LightGBM) benefit from their natural compatibility with TreeSHAP. A universal "best" model cannot be declared globally due to heterogeneous evaluation standards.	Performance comparisons across the literature remain fragile due to mixed target variables, varying units, and inconsistent or absent normalization bases. Consequently, claims of absolute model superiority are rarely portable across studies.
RQ2	SHAP is the dominant global attribution explainer, followed by LIME for localized instance troubleshooting. Permutation Feature Importance (PFI) remains popular for efficient global ranking. Attention visualization is increasingly used but must be strictly distinguished from attention used purely for architectural accuracy.	Quantitative evaluation of XAI quality (e.g., fidelity, robustness, stability) is exceedingly rare. Furthermore, the computational latency of generating explanations (e.g., KernelSHAP on deep models) is frequently omitted, masking deployment feasibility.
RQ3	Across targets and geographic regions, explanations consistently converge on solar irradiance variables as the primary predictive drivers. Temporal encodings (hour, month) and ambient temperature consistently act as critical secondary drivers due to thermal derating effects and cyclical inertia.	Feature rankings are highly conditional on local Köppen climate zones and specific seasons. Global variable rankings should not be treated as universal, as secondary drivers (like humidity and wind) exhibit pronounced seasonal shifts.
RQ4	Short-term horizons (1 to 24 h) are the most studied regime (41 explicit horizon evaluations). Ultra-short-term nowcasting (<1 h) relies primarily on sky images paired with visual saliency/attention maps, while day-ahead forecasting leans on numerical weather predictions (NWP) and satellite inputs paired with post-hoc SHAP analysis.	There is an observable modality shift: spatial image cues dictate ultra-short-term accuracy, while numerical irradiance proxies dominate longer horizons. However, cross-horizon interpretability comparisons within single studies remain limited.
RQ5	Geographic coverage is highly concentrated, with China contributing the largest volume of studies, followed by India and South Korea. The literature spans multiple public (NREL, DKASC) and regional datasets, but climate-stratified benchmarking remains incomplete.	Global geographic diversity is a notable gap. There is a critical underrepresentation of localized forecasting studies in Sub-Saharan Africa and South America, limiting the understanding of model transferability across diverse climates.
RQ6	Primary challenges include heterogeneous benchmarking, a lack of quantitative XAI metrics, and the near total absence of human-in-the-loop operational validation. Future research must prioritize target harmonization, intrinsically interpretable (white-box/physics-informed) architectures, and causal/prescriptive forecasting.	Current literature frequently assumes stakeholder trust rather than empirically measuring it. The transition from passive, post-hoc explanations to active, risk-aware control dispatch represents the critical frontier for grid deployment.

same conclusion from different continents: feature rankings and best-performing model configurations are climate dependent. In Australia, the best model differed between Alice Springs (CatBoost based) and Darwin (AdaBoost based), with LIME showing distinct coastal versus inland predictor profiles [38]. In Algeria, the best feature selection strategy differed between arid Ghardaïa and coastal Algiers [87]. On Saudi data, SHAP importance of humidity and wind direction shifted by season [99]. All three studies are high tier. The consistent implication, also supported by the seasonal encoding analysis in [122], is that model and feature choices tuned in one Köppen zone should be re-validated with local XAI analysis before transfer, and that climate stratified benchmarking is a priority gap (Section 4.2.4).

Which conclusions are supported across multiple independent studies? Table 11 grades the recurring findings. The strongest consensus, irradiance primacy in feature attributions, is supported by nearly every study that reports a global ranking, spanning all model families, targets, and regions. The conditionality of secondary drivers and the scarcity of explanation evaluation are each supported by multiple independent high-tier studies. Findings graded emerging rest on one or two studies and are flagged as such wherever they appear in this review.

4. Discussion

4.1. Main takeaways

Table 10 consolidates the answers to the six research questions together with their caveats. The results show rapid growth in XAI for solar forecasting, with most included studies appearing in 2024 and 2025. SHAP is the most frequently used explainer, while LIME and feature importance methods are commonly used for local and global insight, and attention visualizations appear increasingly in Transformer based studies. Explanations tend to highlight irradiance variables, temporal encodings, and temperature as consistent drivers, while secondary drivers vary by site and season. At the same time, quantitative evaluation of explanation reliability and runtime is uncommon, and human centered validation is rarely reported. These conclusions are supported by the sensitivity analysis of the attention screening rule (Section 2.6) and are stable under quality stratification (Section 3.1.4); the known failure modes of the dominant explainers (Section 3.3.5) qualify, but do not overturn, the coarse-grained consensus on physical drivers.

4.2. Research gaps and future directions

Based on the synthesis of the 173 included studies, six research gaps and corresponding future directions are identified to support progress from methodological studies toward deployment in operational power systems.

4.2.1. Beyond post hoc monoculture toward intrinsic design

SHAP dominance creates a methodological monoculture. While theoretically grounded, standard SHAP assumes feature independence and struggles to untangle complex, collinear physical variables such as irradiance, temperature, and cloud cover. The computational burden of KernelSHAP remains prohibitive for high frequency dispatch.

Future research must pivot toward intrinsic white box architectures that do not require post hoc explainers. Frameworks such as Kolmogorov Arnold Networks [13], Multigene Genetic Programming [41], and Physics Informed Neural Networks [90] represent a critical frontier, as they embed thermodynamic laws and transparent mathematical equations directly into the forecasting mechanism (see Fig. 20).

4.2.2. Rigorous XAI evaluation frameworks

The most pressing validation gap is the scarcity of XAI specific evaluation metrics. While pioneering exceptions exist, such as [42] measuring explanation fidelity after dataset compression and [103] validating attribution stability using Kendall's tau, most studies rely on visual inspection rather than mathematical verification.

Future research must establish standardized, domain specific evaluation protocols. Benchmark datasets equipped with ground truth feature attributions derived from deterministic physical clear sky models are urgently needed to objectively score fidelity and robustness of competing XAI outputs.

Table 11
Consistency grading of consolidated findings across independent studies. Quality tiers: H high, M medium, L low, n/s = individual score not available. Grades: Strong ≥4 independent studies including ≥3 high tier, or near universal pattern; Moderate 3–4 studies or mixed tiers; Emerging 1–2 studies.

Consolidated finding	Supporting studies	Quality tiers	Consistency
Irradiance components (GHI, DNI, DHI) are the primary XAI-identified drivers across targets, architectures, and regions	[38,85,87,99,102,103]; 104 studies overall (Table A.1)	H,H,H,H,H,H	Strong
Temporal and cyclical encodings are stable secondary drivers	[86,102,122,132]	H,n/s,H,H	Strong
Ambient temperature is a persistent secondary driver consistent with thermal derating	[74,86,99,102]	H,H,H,H	Strong
Secondary drivers (humidity, wind, cloud proxies) are season- and climate-conditional; universal feature rankings are not supported	[38,87,99,122]	H,H,H,n/s	Strong
No architecture family shows portable cross-study superiority; the best-performing model within a study is typically the proposing hybrid	Sections 3.2, 3.5.1	corpus-wide	Strong (negative finding)
Tree ensembles with TreeSHAP give exact, low-cost attribution for tabular targets	[31,32,85,86,115]	H,H,H,H	Strong
Sub-hour horizons favor sky-image models with visual explanations; accuracy degrades under broken clouds	[26,76–78]	M,M,H,M	Moderate
Attention and TFT models provide built-in variable selection usable as importance	[80,81,83,84]	M,H,M,L	Moderate
SHAP runtime is a deployment bottleneck; clustering or compression mitigates it	[42,59]	n/s,H	Emerging
Quantitative evaluation of explanation quality (fidelity, stability) is feasible but rare	[42,102,103] vs. 173-study corpus	H,H,n/s	Strong (gap)

4.2.3. Real time deployment in distributed energy resources

For operational energy management, the trade off between explanation quality and runtime remains a key bottleneck. Runtime reporting is limited, and several studies indicate SHAP can be slow for high frequency control loops.

Future investigations must prioritize approximate attribution methods, knowledge distillation, and clustering techniques such as [42]. Most importantly, the field must transition from passive explanation to active control. The SHAP EMS paradigm [59], where real time XAI outputs are fed directly into rule based controllers to dynamically adjust residential battery dispatch, represents a promising blueprint for autonomous distributed energy systems, particularly for risk-based scheduling [143], green hydrogen optimization [144], and environments with highly uncertain, limited data [131].

4.2.4. Climate stratified and spatiotemporal analysis

The location dependent XAI findings of [38] across Australian zones, coupled with seasonal variability documented by [99], demonstrate that universal feature rankings are physically invalid.

Future research must embrace climate stratified benchmarking. Systematic multi site XAI analyses mapped across diverse Köppen climate zones, particularly incorporating data from underrepresented regions such as Sub Saharan Africa and South America, are required to establish globally transferable forecasting guidelines.

4.2.5. Human in the loop validation and LLM integration

The practical value of explainability is assumed but rarely tested on actual grid operators. Controlled behavioral experiments comparing dispatch operator performance with and without XAI tools are essential.

Integrating Large Language Models with XAI outputs offers a promising direction. Early architectures such as SolarTrans [71] and XGBoost GPT [69] demonstrate that LLM agents can translate complex numerical SHAP arrays into natural language weather narratives, lowering the cognitive barrier for system operators (see Figs. 20 and 21).

4.2.6. The causal and prescriptive frontier

The literature is overwhelmingly focused on correlational attribution, while causal inference and prescriptive planning remain underexplored. To move from predictive to prescriptive analytics, future frameworks must adopt Diverse Counterfactual Explanations [114] to provide operators with actionable what if scenario planning.

Advancing the mathematical foundations of Causal Machine Learning, such as the A PDD SHAP algorithm [106] which injects Directed Acyclic Graphs to force models to attribute importance to root physical causes rather than statistical proxies, will define the next generation of solar forecasting.

4.3. The reproducibility gap and its consequences

The quality assessment exposes a structural weakness that deserves separate discussion: only 3 of 173 included studies (1.7%) received a full reproducibility score by making their code and data openly available (Q10), 61.3% provide neither, and the remainder offer partial statements such as data available on request. This gap is uniform across quality tiers (Fig. 12b), so it is a disciplinary norm rather than a property of methodologically weaker studies. Its consequences compound in an XAI review in at least four ways.

First, performance claims cannot be independently verified. Because most studies use private plant data with undocumented preprocessing, the reported error metrics cannot be verified or refuted by third parties, and the benchmarking fragility documented in Section 3.5.1 cannot be corrected retrospectively by readers or reviewers. Second, and specific to explainability, explanation outputs are even more implementation sensitive than error metrics: SHAP values depend on the estimator variant and background dataset, LIME on its sampling configuration, and saliency maps on the layer and baseline choices. Without code, a published feature ranking cannot be regenerated even when the data are public. Third, the gap inflates apparent novelty and slows convergence. New hybrids are compared against re-implemented baselines of unverifiable fidelity, which helps explain why the best-performing models within individual studies rarely replicate across studies. Fourth, it weakens the field's own transparency argument: a literature that holds that forecasts must be transparent to be trusted currently rests

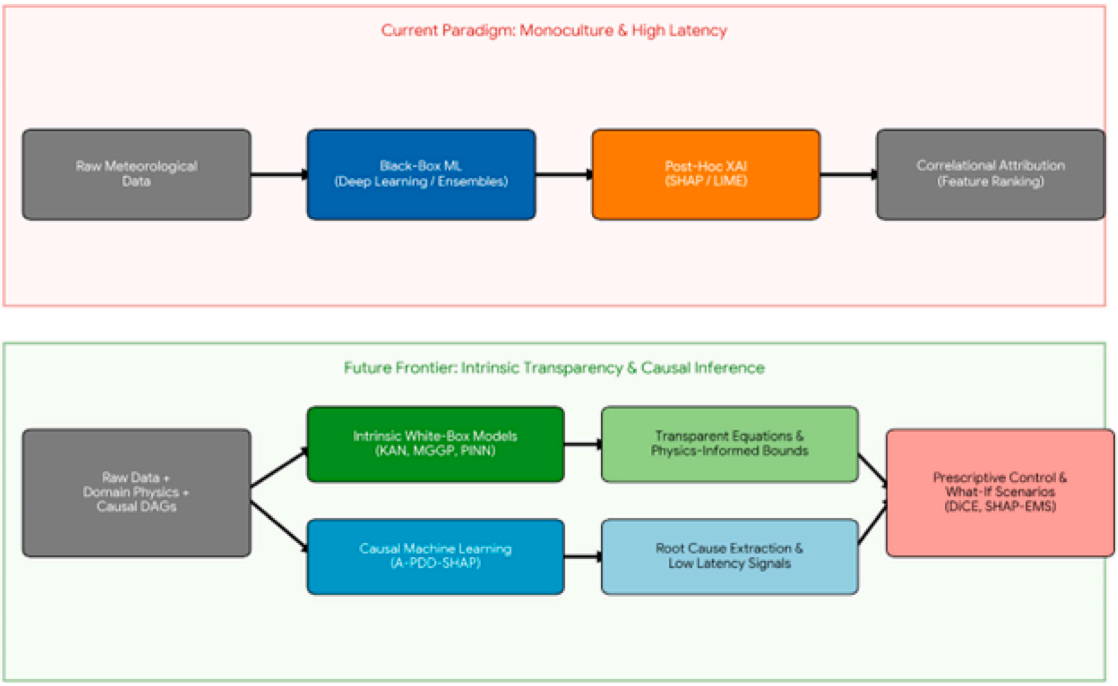


Fig. 20. Proposed methodological shift in solar forecasting.

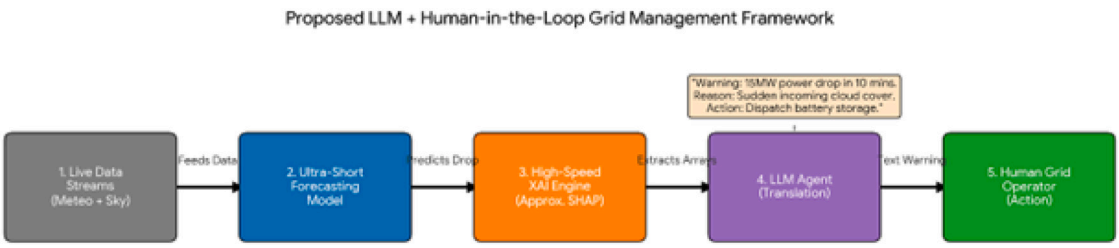


Fig. 21. Proposed operational framework integrating real-time XAI.

on research artifacts that cannot themselves be inspected, a tension that editors, referees, and funders are in a position to resolve.

Practical remedies exist and are inexpensive relative to the modeling effort these studies already invest. We recommend, in order of expected impact: mandatory code and configuration release for the explanation pipeline, not only the forecasting model; preferential use of public reference datasets (NREL NSRDB and SRRL, DKASC, Stanford sky image sets) at least as secondary benchmarks; explicit reporting of the XAI implementation details listed above; and adoption of explanation-specific reporting checklists as a submission requirement. Until such norms are established, unreleased single-site results, including their explanations, should be treated as provisional evidence.

4.4. Limitations

This systematic review has limitations. The search was restricted to English language publications, which may exclude relevant work published in other languages. The search was also limited to four databases, Scopus, Web of Science, IEEE Xplore, and ScienceDirect, so some niche venues may be missed. The sensitivity analysis of the EC2 screening rule (Section 2.6) re-screened a 25% random sample of the excluded records at title and abstract level; although the estimated false exclusion rate was zero with a narrow confidence interval, a paper that applied XAI at full text without signaling it in the title, abstract, or keywords would evade both the original screening and the re-screening, so a small residual misclassification risk remains. Quality scores were assigned with a single reviewer-team consensus procedure rather than fully independent double scoring, and any quality checklist compresses methodological nuance into ordinal categories. Finally, the included literature spans different targets and units, and many studies do not report a clear normalization basis, which limits direct comparability even when synthesis is stratified by target class. For this reason, all cross-model performance statements in this review should be read as within-study evidence or as conditional tendencies, not as global rankings.

5. Conclusions

This paper reports a PRISMA systematic review on explainable artificial intelligence for solar forecasting, with a focus on interpretability as an implemented analytic output rather than a label. From 951 records retrieved across Scopus, Web of Science, IEEE Xplore, and ScienceDirect, we removed duplicates and applied eligibility checks that verify genuine explanation generation and quantitative forecasting evaluation, yielding 173 studies for full text synthesis. A central methodological contribution is the separation of attention used only to boost predictive accuracy from attention that is actually inspected as an explanation, this boundary excluded 299 papers. A dedicated sensitivity analysis supports this rule: none of the 299 excluded records mentions a recognized explanation technique, and dual independent re-screening of a 25% random sample found no false exclusions (95% CI 0–4.9%). A quality-stratified synthesis further shows that the principal rankings

Table A.1
Consensus Input Feature Categories, Physical Roles, and Literature Coverage. The frequency column counts studies where XAI analysis identified the category as an influential driver; the coverage column lists all studies reporting the feature category, and is therefore broader for some rows.

Feature category	Studies frequency	Physical proxy/Predictive function	Extensive study coverage
Solar Irradiance (GHI, DNI, DHI, Clearness)	104	Direct energy source; defines atmospheric scattering and transmittance effects.	[5,8,14,15,25,26,29,31–33,39,43,44,47,48, 50,53,55,58–60,62,64,65,68,70,71,75–77, 80–82,84,86,90,91,95–99,101–103,106–111, 121,123,124,126,127,132,135,137,139,141, 145–188]
Temporal & Cyclical Encodings (Hour, Day, Month, Season)	89	Earth-Sun geometry; captures diurnal and seasonal inertia proxies.	[8,9,12,14,15,25,26,29,30,32,38,39,41,43, 45,47,48,50,53,57,58,62,65,68–71,73–76,78, 79,81,82,85,86,91,92,94–97,99,102,103,106, 107,110,111,114,123–125,132,135–139,142, 146–151,153,155,156,158,159,161,162,164– 166,168–173,176–178,182–184,187–202]
Temperature (Ambient, Module, Soil)	54	Hardware efficiency degradation (thermal derating) and environmental thermal inertia.	[5,26,32,33,45,54,56,58,60,62,64,70,71,75, 84,86,90–92,95–97,99,101,107–109,123, 124,127,137,145,146,148,149,152,153,159, 168–172,174,177,180,183,186–188,194,199, 203,204]
Cloud Cover & Sky Indices	37	Cloud opacity, shading limits, and localized atmospheric thickness.	[15,29,31,32,41,43,44,47,53,68,76–78,81, 103,106,127,128,139,151,154–158,160,161, 164,167,169–171,185,193,195,198,203]
Relative Humidity	36	Atmospheric transmittance, vapor density, and moisture scattering.	[5,26,32,33,54,56,60,62,70,90,91,95,96,99, 102,107,111,123,124,127,137,153,154,164, 168,170–172,174,177,179,180,183,187,203, 204]
Wind Speed & Direction	20	Convective cooling of PV panels and weather front movement tracking.	[26,32,33,56,60,62,99,107,123,124,137,148, 168–172,178,186,205]
Precipitation & Pressure	12	Barometric shifts signaling rapid storm arrival and panel soiling wash-off.	[5,26,32,33,54,60,86,90,92,94,123,138,164, 170–172,177,183,188,203]
Linguistic/Categorical Text	3	Human-readable cloud/weather observations translated into structured probabilities.	[167,171,198]

are unchanged when the corpus is restricted to the 114 high-quality studies, so the conclusions below are not driven by low-quality studies.

Across target classes that include irradiance components, time integrated solar radiation, and PV power output, the literature shows rapid growth and mixed reporting discipline. Most included studies were published in 2024 and 2025, interpretability is no longer a side note in solar forecasting work. SHAP is the most widely adopted explainer with 69 studies, while LIME appears less often yet remains valuable for local diagnostics, and 27% of the corpus combines two or more explainers. Explanations converge on physically plausible drivers, irradiance variables remain primary, temporal encodings are consistently strong, temperature follows as a persistent efficiency proxy. Secondary drivers such as humidity, wind, and cloud indicators shift with season, site, and climate zone, so global feature rankings should be treated as conditional not universal, this nuance is still underreported.

Even with strong adoption, the field is not ready for robust operational trust claims. Cross study benchmarking is fragile because targets, units, horizons, and normalization bases remain heterogeneous, and explanation quality is rarely measured using fidelity, stability, sensitivity, or runtime reporting. Reproducibility infrastructure is likewise underdeveloped, with fewer than 2% of studies releasing both code and data. Human centered validation with grid operators is also mostly absent, so the practical value of explanations is assumed rather than demonstrated. Moving forward, trustworthy solar forecasting will require target harmonization, standardized quantitative XAI evaluation, and more intrinsically interpretable or physics informed designs, paired with causal reasoning and prescriptive counterfactual analysis. These steps can turn explanations into decision support that is auditable, fast enough for deployment, and credible under real grid constraints.

AI tools statement

Generative AI assistants (Gemini, the Overleaf built-in AI assistant, and Claude) were used to assist with grammar, spelling, vocabulary, and editorial polishing of this manuscript. All scientific content, screening decisions, analyses, and conclusions were produced and verified by the author, who take full responsibility for the content of the publication.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. Supplementary tables

See Tables A.1 and A.2.

Data availability

No new experimental data were generated. The PRISMA screening records (including the full list of 299 EC2-excluded records and the sensitivity-analysis sample), the database query strings, the 15-field extraction dataset, and the per-study quality assessment scores are available from the corresponding author on request.

Table A.2
Performance metric ranges by target variable across solar forecasting studies, best reported per paper.

Target variable	Algorithm families	RMSE	MAE	nRMSE	R^2	MAPE	MBE	SS	Acc
GHI Global horizontal irradiance	LSTM, BiLSTM, GRU CNN, TCN Transformer, TFT XGBoost, LightGBM, CatBoost Random Forest Hybrid	15-180 $W\,m^{-2}$	8-120 $W\,m^{-2}$	5%–35%	0.82–0.99	3%–25%	−15 to +20 $W\,m^{-2}$	0.15–0.65	–
PV power output	LSTM, BiLSTM, GRU CNN-LSTM hybrids Transformer, TFT XGBoost, LightGBM, CatBoost Random Forest Hybrid, physics informed	0.5-50 kW 3%–20% norm.	0.3-30 kW 2%–15% norm.	4%–25%	0.85–0.99	2%–20%	−5 to +5 kW	–	–
DNI Direct normal irradiance	LSTM, GRU Transformer XGBoost, LightGBM Random Forest Hybrid	30-250 $W\,m^{-2}$	20-150 $W\,m^{-2}$	12%–45%	0.70–0.95	–	–	–	–
DHI Diffuse horizontal irradiance	Boosted trees LSTM Hybrid	20-100 $W\,m^{-2}$	10-70 $W\,m^{-2}$	–	0.75–0.93	–	–	–	–
Solar radiation time integrated daily or monthly	ANN, MLP SVR, ELM Random Forest, XGBoost, LightGBM Regression hybrids	0.5-5.0 $MJ\,m^{-2}\,day^{-1}$ or $kWh\,m^{-2}$	0.3-3.0 $MJ\,m^{-2}\,day^{-1}$	–	0.88–0.99	2%–12%	–	–	–
Cloud cover transmittance sky image nowcasting	CNN, ResNet, ConvLSTM Attention vision models Multimodal CNN + tabular ML	40-200 $W\,m^{-2}$	25-130 $W\,m^{-2}$	–	–	–	–	0.10–0.55	85%–97%
Clearness index K_t and normalized proxies	XGBoost, Random Forest, SVR LSTM Lightweight regressors	0.05–0.20	0.03–0.15	–	0.80–0.96	–	–	–	–

Note: Irradiance is instantaneous W/m^2, solar radiation is time integrated energy such as $MJ/m^2\,day^{-1}$ or $kWh/m^2\,day^{-1}$, PV power is W, kW, or MW.

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